Responsible Agentic Reasoning via AI Agents: A Critical Survey of Agentic AI

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Abstract

Responsible use of trustworthy Artificial Intelligence (AI) is entering a pivotal stage, where Large Language Model (LLM)-based agents excel at integrating multi-source knowledge into coherent reasoning chains. However, these agents remain opaque and difficult to audit in the absence of embedded, in-loop safety mechanisms. Existing surveys treat reasoning, agentic behavior, and safety in isolation, leaving a gap in how to integrate them into practical, trustworthy agents. To address this, we present a survey at the intersection of these domains and introduce **Responsible Reasoning AI Agents ($\mathbf{R^2A^2}$)**, a class of agentic LLM systems that generate explicit reasoning traces while enforcing fairness, privacy, transparency, accountability, and auditability throughout the decision loop. We synthesize recent advances in chain-of-thought prompting, ReAct, tree/graph-of-thought structures, tool use, memory, retrieval, and agentic browsing, and integrate these with responsible AI principles into a unified evaluation framework. Furthermore, we propose an evaluation methodology for agentic reasoning with embedded safety mechanisms and outline a five-stage reproducible protocol: Curate, Unify, Probe, Benchmark, and Analyze; to operationalize responsibility metrics. This taxonomy, metric suite, and framework advances the development of safe, transparent, and governable LLM-based agentic AI systems. The project repository is publicly available at GitHub.

Keywords: Responsible AI; Agentic AI; AI Agents; Agentic Reasoning; Explainable AI

1. Introduction

Reasoning - the process of turning facts into defensible decisions - is rapidly emerging as a defining capability of next-generation AI models. The community increasingly views reasoning as the ability to draw justified conclusions from facts, enabled by deeper representations and part–whole hierarchies [1]. Recent Large Language Models (LLMs) already solve logic puzzles, plan experiments, and write executable code by chaining "think-then-act" steps [2]. When deployed as agents [3], they call external tools, store intermediate thoughts, and interact with complex environments.

At its core, every agentic reasoning step is an act of multi-stage *information fusion*: the agent must combine a user query, retrieved documents, tool outputs, prior reasoning states, and policy constraints into a single, coherent next action [4]. Classical information fusion taxonomies distinguish data-level, feature-level, and decision-level integration [5]; LLM-based agents instantiate all three. Retrieval-augmented generation (RAG) performs data-level fusion by injecting external evidence into the context window. Internal attention mechanisms realize feature-level fusion across modalities and reasoning traces [6]. And the agent plan-execute- validate loop enacts decision-level fusion, aggregating heterogeneous signals (correctness, safety, privacy) into a final output [7]. Despite this deep connection, existing surveys rarely frame agentic reasoning through the lens of such multi-stage information and reasoning fusion, missing opportunities to leverage decades of fusion-theoretic work on uncertainty propagation, conflict resolution, and quality assessment. This survey explicitly adopts an information-fusion perspective, positioning responsible reasoning as the *quality control layer* of a multi-source fusion pipeline.

In parallel, the underlying language models followed a similar developmental trajectory, for example, the early transformer-based models (e.g., BERT, GPT-3; 2018-2020) achieved strong pattern recognition but often struggled with systematic multi-step reasoning without additional scaffolding. Empirical studies suggest that scaling parameters alone was often insufficient to reliably support multi-step deduction or abstraction [8]. Chain-of-thought (CoT) prompting (2022) introduced explicit intermediate reasoning traces that substantially improved arithmetic and symbolic benchmarks [9], while ReAct [10] and Tree-of-Thought [11] extended this paradigm through tool-mediated interaction and branching search, advancing the field toward LLM-based agentic planning. Although early

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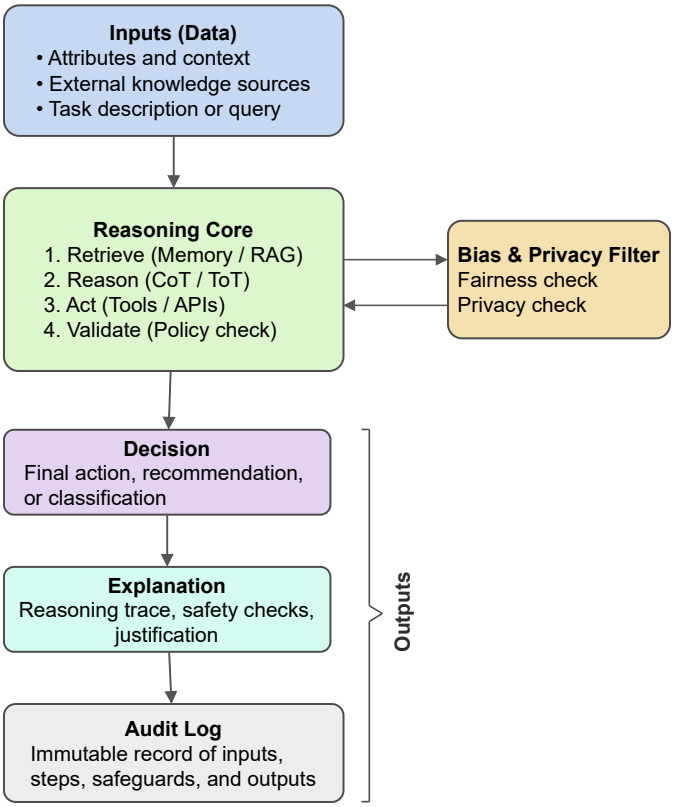


Figure 1: Example of responsible reasoning pipeline. Data inputs are processed by a Reasoning Core (retrieve, reason, act, validate) that interfaces bidirectionally with a Bias & Privacy Filter. The pipeline produces three outputs: a Decision, an Explanation, and an immutable Audit Log

agent frameworks (e.g., LangChain, AutoGPT) operationalized these capabilities, where safety was frequently implemented via external guardrails rather than being intrinsically embedded within the agent's reasoning loop. More recent work, particularly in the context of LLM-driven agents, increasingly introduces benchmarks that evaluate not only final outputs but also intermediate reasoning traces for properties such as faithfulness, safety, and privacy [12]. This progression motivates the present survey. We introduce **Responsible Reasoning AI Agents ($\mathbf{R^2A^2}$)** , a unifying evaluation and design paradigm that reconceptualizes agentic reasoning as a governed information-fusion process with trace-level accountability.

Related Surveys and Gaps in the Literature. Recent surveys have substantially advanced our understanding of LLM reasoning and responsible AI, but these strands are often reviewed in isolation. Surveys such as [13, 2] provide taxonomies of reasoning strategies (e.g., CoT, ToT, ReAct), while complementary work surveys fairness, safety, and risk mitigation for LLMs [14, 16]. In parallel, surveys on agentic architectures and evaluation [15, 3, 17, 18, 28] examine planning, memory, tool use, and trace- or trajectory-based assessment. Although some works [4, 29] emphasize advances in reasoning performance and agent design, fewer surveys explicitly treat *responsibility as an in-loop property* of multi-step reasoning. Our survey addresses this gap by synthesizing methods that embed and assess safeguards: including fairness, privacy, transparency, and auditability, within the agent decision loop, at the level of intermediate reasoning traces and tool-mediated actions. Table 1 summarizes prior surveys and positions our R^2A^2 review at the intersection of reasoning, agency, and responsibility.

Relationship to Existing Frameworks. R^2A^2 is also complementary to orchestration frameworks such as LangChain [30], LlamaIndex [2], and AutoGen [31]. These toolkits provide practical infrastructure for agent construction (e.g., tool routing, retrieval augmentation, and memory management). R^2A^2, by contrast, provides an evaluation and accountability lens: it specifies trace-level responsibility properties and associated metrics for auditing the agent's reasoning process. In principle, agents implemented in any orchestration framework can be assessed under R^2A^2, provided they expose structured traces of reasoning steps, tool calls, and observations.

Survey Contributions. This survey synthesizes a broad body of recent literature and provides: (i) a unified taxonomy spanning reasoning methods, agent components (tools, memory, retrieval), and in-loop safeguards; (ii) a reproducible evaluation protocol (*Curate, Unify, Probe, Benchmark, Analyze*) for scoring both task success and trace-level responsibility; and (iii) operational metrics (e.g., RRI, CoT-RS, RRC, AGR) computable from standard agent traces to surface correctness, transparency, and governance signals beyond answer-only evaluation. Additionally, we provide (iv) an explicit information-fusion framing that connects agentic reasoning to classical fusion-theoretic concepts (source reliability, conflict detection, quality assessment) to provide cross-pollination between the information-fusion and agentic-AI communities. An example of a responsible reasoning pipeline is shown in Figure 1, where the goal is to integrate fairness, privacy, and auditability within the retrieve - reason - act - validate loop, enabling trace-level accountability throughout the decision process. This pipeline makes explicit the coupling between reasoning steps and responsibility checks, enabling trace-level auditing.

Review Scope and Bibliometric Overview. This survey adopts a structured, narrative review informed by scoping-review best practices [32, 33]. Our scope spans three intersecting domains: (i) reasoning in LLMs, (ii) responsible AI, including fairness, transparency, safety, and privacy, and (iii) agentic AI systems. We considered studies published from January 2024 to January 2026, along with a small set of foundational papers published prior to 2023. We collected papers from major peer-reviewed academic sources, including the ACL Anthology, IEEE Xplore, SpringerLink, DBLP, Scopus, Web of Science, and additionally incorporated preprints from arXiv.

During search, the example keyword queries combined Boolean operators and phrase matching to ensure broad

[2] LlamaIndex

Table 1: Comparison of major surveys on LLM reasoning, responsible AI, and agentic architectures. ✓ = primarily addressed, △ = partially addressed, – = not addressed.

Survey	Multi-step reasoning	Responsible AI (fairness / safety)	Agentic LLM agents	In-loop safeguards	Trace-level evaluation
CoT Reasoning [13]	✓	△	-	-	-
Fair LLMs [14]	-	✓	-	-	-
LLM Reasoning [2]	✓	△	△	-	△
Agentic LLMs [15], [3]	✓	△	✓	-	△
Responsible LLMs [16]	△	✓	△	△	△
Agent Evaluation [17]	△	△	✓	-	△
Reasoning Trace Evaluation [18]	✓	△	-	-	✓
RLM Blueprint [19]	✓	△	△	-	△
100 Days After DeepSeek-R1 [20]	✓	✓	△	△	△
TRiSM for Agentic AI [21]	△	✓	✓	△	△
Agentic AI Security [22]	△	✓	✓	△	△
Large Reasoning Models [23]	✓	△	△	-	△
Multi-Agent Risks [24]	-	✓	✓	△	-
Holistic Agent Leaderboard [25]	△	△	✓	-	✓
Deep Research Agents [26]	✓	△	✓	-	△
Reasoning Agents [4]	✓	✓	✓	△	△
LM Agent Cooperation [27]	△	✓	✓	△	-
Internet of Agents [28]	-	✓	✓	△	-
This survey (R^2A^2, 2026)	✓	✓	✓	✓	✓

yet targeted retrieval. For reasoning in LLMs, we used queries such as "reasoning" AND "language model", "chain-of-thought" OR "tree-of-thought" OR "graph-of-thought", and "reasoning strategies" AND ("prompting" OR "search"). For responsible AI, we included terms such as "responsible AI" AND ("fairness" OR "bias mitigation" OR "transparency" OR "explainability" OR "safety" OR "privacy"), as well as governance-oriented phrases including "ethical AI", "AI governance", and "trustworthy AI". For agentic systems, we searched "LLM agent", "agentic LLM", "autonomous language agent", and ("multi-agent system" OR "agent orchestration") AND ("safety" OR "responsibility"). Evaluation-focused queries included "reasoning trace" AND ("evaluation" OR "auditing" OR "faithfulness"), and "TRiSM" AND ("agents" OR "agentic systems"). Queries were iteratively refined to balance recall and precision, ensuring comprehensive coverage across reasoning, responsibility, and agentic AI domains. Two independent reviewers screened titles and abstracts, followed by full-text assessment. Disagreements were resolved through discussion and consensus.

Figure 2 synthesizes the bibliometric landscape of the R^2A^2 corpus across four complementary dimensions. Panel (a) reveals a sharp inflection in publication volume after 2023, rising from 47 papers in 2023 to 231 in 2025, reflecting the rapid transition into the LLM-driven agentic era (the modest 2026 count of 10 is an artifact of the January cutoff). Panel (b) shows that LLMs, AI agents, and reasoning collectively dominate the thematic distribution, while governance and evaluation oriented topics, including benchmarks, bias and fairness, responsible AI, and AI safety, form a substantial and growing cluster accounting for roughly one third of all entries. The topic–year heatmap in Panel (c) further disaggregates these trends: AI agents and reasoning exhibit the steepest year-over-year growth, from 12 to 76 and 9 to 52 entries respectively between 2023 and 2025, whereas responsible AI and bias/fairness themes, though smaller in absolute volume, appear almost exclusively from 2024 onward, signaling that responsibility-aware research is a recent but accelerating strand. Finally, Panel (d) highlights the publication venue distribution: arXiv dominates with 150 entries, consistent with the field's rapid preprint-driven pace, while peer-reviewed outlets such as Springer, NeurIPS/ML venues, IEEE, and ACM collectively contribute a balanced complement, indicating that the literature is beginning to mature beyond preprints into established conferences and journals. Taken together, these trends underscore a field-wide pivot from foundational model scaling toward structured reasoning, systematic evaluation, and responsibility-aware deployment, precisely the intersection that motivates the R^2A^2 framework.

Paper Organization: The remainder of this paper is organized as follows. Section 2 introduces the foundations of LLM-based reasoning and responsible AI. Section 3 reviews reasoning methods, responsible mechanisms, and agentic architectures. Section 4 presents ourproposed unified evaluation framework for R^2A^2. Section 5 presents the modular design for implementing R^2A^2. Section 6 summarizes gaps, future directions, and Section 7 is the conclusion.

2. Foundations of Reasoning Language Agents

In this section, we outline core concepts and terminology used throughout the R^2A^2 survey.

2.1. Background of Agents

AI agents are software systems that perceive their environment, reason about inputs, and execute actions to achieve specified goals [44]. Common categories include: (1) *autonomous agents* that operate without continuous human oversight [57]; (2) *multi-agent systems* in which

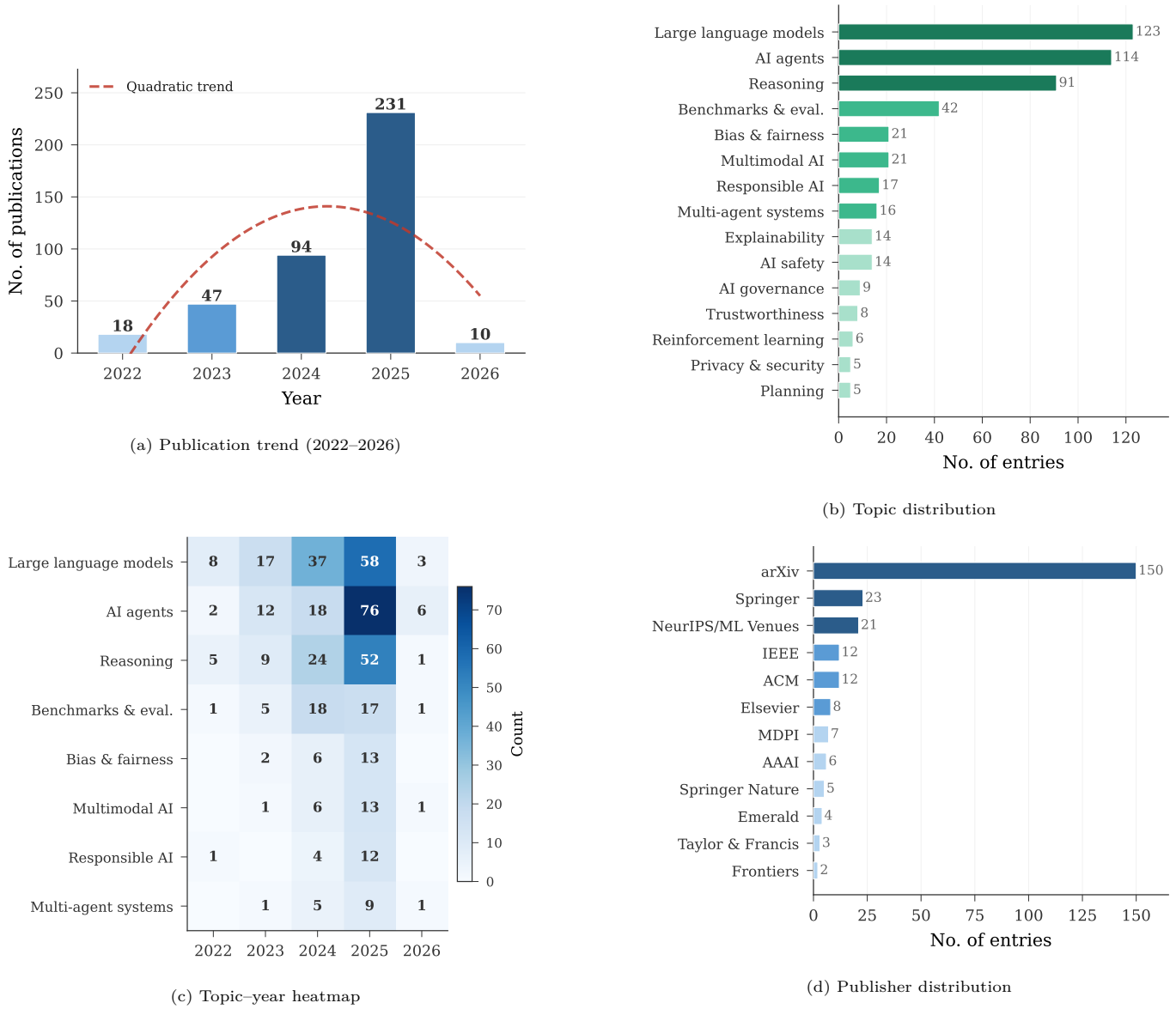


Figure 2: Bibliometric analysis of papers (2022–2026): (a) annual publication count, (b) topics' frequency, (c) topic–year heatmap for the top research themes, and (d) publisher distribution.

multiple agents coordinate or compete [58, 27]; and (3) *LLM-based agents* that use large language models for planning, reasoning, memory, and tool use [16].

Agent architectures have evolved from symbolic and rule-based systems [59], to reactive approaches and deliberative frameworks such as BDI (belief-desire-intention) [36], and further to learning-enabled, multimodal, and LLM-powered agents [60]. Modern LLM agents often combine structured reasoning (e.g., CoT/ToT/ReAct) with long-term memory, retrieval, and external tool access [9, 11, 10]. Recent work also explores large-scale, multi-agent interaction settings (e.g., web- or internet-enabled coordination) where heterogeneous agents communicate and collaborate across tasks [28]. Building on prior taxonomies, Table 2 organizes contemporary AI agents along five types: degree of autonomy, underlying architecture, interaction paradigm, intelligence backbone, and primary functional role.

2.2. Evolution From LLMs to Reasoning AI Agents

Early transformer-based models (e.g., BERT, GPT-3; 2018 - 2020) achieved strong pattern learning but often struggled with systematic multi-step reasoning without additional scaffolding. As research progressed, it became increasingly clear that parameter scaling alone was not sufficient to *reliably* yield robust multi-step deduction or abstraction across settings [8]. This motivated a shift toward *reasoning-centric* methods and agentic systems that structure intermediate steps, modularize cognition, and expose traces for inspection [61].

The introduction of CoT prompting (2022) enabled models to produce explicit intermediate steps, substantially

Table 2: Types of AI agents.

AI Agents Terms	
Category	**Representative Forms**
Autonomy [34]	Human-in-the-loop [35]; Mixed-initiative; Fully autonomous [34].
Architecture	Deliberative (BDI) [36]; Reactive [37, 38]; Hybrid [39, 40]; Cognitive architectures (Soar/ACT-R, Active Inference).
Interaction [41]	Single-agent [42]; Multi-agent [3] (cooperative, competitive, role/protocol); Human–AI collaboration.
Intelligence [43]	Rule-based / Symbolic [44]; ML-based (RL/SL) [45]; LLM-based [46]; Neuro-symbolic; Planning & Tool Use [10].
Function [43, 3]	Goal-driven / Task agents; Conversational [47]; Tool-augmented [10]; Embodied / Robotics; Research / Science agents.
Memory & Knowledge [48, 6, 49]	Short-term (scratchpad) [9]; State/Episodic [48]; Long-term (RAG/vector DB) [6, 50]; Semantic/KG; Procedural skills.
Cognition Agents [51]	Perception–Cognition–Action loop [52]; Working memory & attention [53]; Reasoning (model-based, model-free [45], hybrid/meta-control); Metacognition/self-reflection [54]; Self-model / Theory of Mind [55]; Norms & value alignment [56].

improving performance on arithmetic and symbolic reasoning benchmarks [9, 23]. In 2023, ReAct [10] and Tree-of-Thought (ToT) [11] extended CoT through tool-mediated interaction and branching search, advancing the field toward agentic planning. Open-source frameworks such as LangChain [30] and AutoGPT [62] operationalized these ideas in practical pipelines; however, safety controls were often implemented as external guardrails rather than being intrinsic to the agent's reasoning policy.

Beyond prompting, reinforcement learning (RL) can optimize multi-step decision trajectories through iterative feedback [45], and hybrid neuro-symbolic systems combine neural generalization with symbolic precision [76]. Techniques such as Monte Carlo Tree Search (MCTS) [77], agent-to-agent orchestration [78], and retrieval-augmented generation (RAG) [79] further enable exploration, verification, and adaptation of reasoning paths. Enterprise platforms (e.g., Amazon Bedrock AgentCore [80], Microsoft Copilot Studio [81]) increasingly support secure tool use, role specialization, and auditable workflows at scale.

More recent work (2024-2026) increasingly explores *in-loop* governance, where safeguards (e.g., policy checks, privacy controls, and trace logging) are integrated into the agent step-by-step process rather than applied only after generation. Concurrent benchmarks evaluate not only final answers but also intermediate reasoning traces for properties such as faithfulness, safety, and privacy. Figure 3 summarizes this progression from LLMs to reasoning-centric and responsibility-aware agent frameworks.

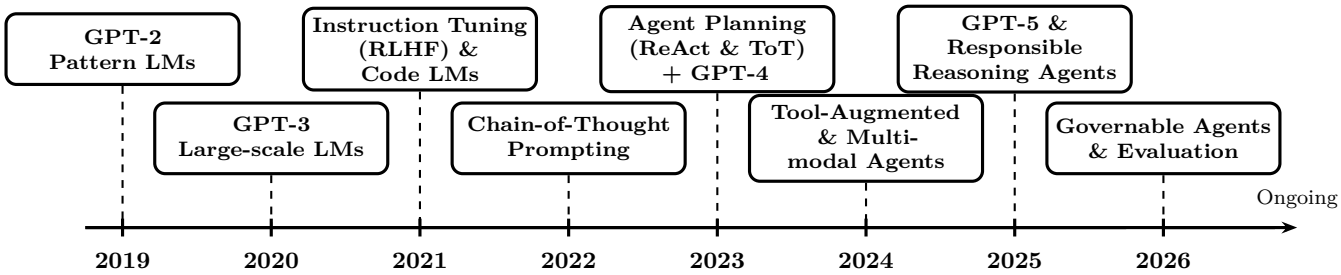
2.3. Defining Responsible Reasoning AI Agents (R^2A^2)

We define an R^2A^2 agent as an LLM-based agentic system that performs multi-step reasoning and exposes structured intermediate traces (e.g., steps, tool calls, observations) to support verification and evaluation. Each step may invoke symbolic operations, retrieval, or external tools, and can be subject to built-in policy checks for bias, privacy, and auditability. Unlike conventional pipelines where safeguards are applied primarily as post-hoc filters, R^2A^2 emphasizes *in-loop* responsibility: intermediate steps are monitored and evaluated, not only the final output. This design supports trace-level inspection and auditing, which is especially important in high-stakes settings where accountability and governance are required. A visual overview is shown in Figure 4.

2.4. Basic Components of R^2A^2

We summarize the basic components of R^2A^2 as follows:

- **Reasoning schemes:** R^2A^2 agents can represent intermediate steps as chains, trees, or graphs [82]. CoT supports linear deduction (e.g., arithmetic reasoning) [9], while tree/graph structures enable parallel exploration of alternative hypotheses (e.g., clinical decision-making) [83]. These structures improve traceability, enabling human auditors to verify or correct the agent's logic [84, 85].
- **Tree-based reasoning (ToT):** ToT branches over candidate partial solutions [11]. It defines the search space; algorithms such as Monte Carlo Tree Search (MCTS) or beam search can then navigate this space to select high-quality solutions.
- **Neural operators:** Policy/value/critic modules can steer step selection and estimate utility [86]. When used, these components are trained with gradient-based methods; responsibility constraints can be incorporated via objectives and/or step-level policy checks.
- **Memory and retrieval:** R^2A^2 agents may use an ephemeral scratchpad, episodic context buffers, and vector-indexed long-term stores for grounding [79]. Retrieval augmentation can improve factuality without increasing model parameters.
- **Policy and safety enforcement:** A policy engine can validate each step against bias, privacy, and security constraints using rule-based filters and/or learned reward models [84, 87].
- **Audit logging:** An append-only, tamper-evident log can record reasoning steps, tool interactions, and outputs to support *post hoc* compliance checks and oversight [84].
- **Tool and execution:** Secure APIs connect agents to external tools (e.g., calculators, databases, browsers) for real-world actions [88]. Sandbox execution and least-privilege design mitigate unintended side effects.



Year	Milestone	Key Models/Advancements (examples)
2019	Pattern LMs	GPT-2 (OpenAI) [63]
2020	Large-scale LMs	GPT-3 (OpenAI) [64]
2021	Instruction tuning & Code LMs	InstructGPT (RLHF) [65]; Codex [66]
2022	Chain-of-Thought Prompting	CoT [9]
2023	Agent Planning + GPT-4	ReAct [10], ToT [11], GPT-4 [63]
2024	Tool-Augmented & Multimodal Agents	GPT-4o [63]; Gemini 1.5 Pro [67]; Claude 3 family [68]; Llama 3/3.1 [69]
2025	GPT-5 & Responsible Reasoning Agents	GPT-5 (Aug 2025) [63]; OpenAI o3 / o4-mini [63]; GPT-5.1 Thinking [63]; GPT-5.2 [63]; Gemini 2.5 Pro [67]; Gemini 2.5 Deep Think [67]; Gemini 3 Pro [67]; Claude Sonnet 4 [68]; Claude Opus 4.5 [68]; Llama 4 [70]; Grok 3 [71]; IBM Granite 3.2/3.3 [72]; ERNIE 4.5 [73]; Mistral Medium 3 / Magistral [74], [75]
2026	Governable Agents & Evaluation	Claude Opus 4.6 [68]; GPT-5.3/5.4 [63]; EU AI Act enforcement (Feb 2026); R^2A^2 framework (this work)

Figure 3: Milestones and timeline of language model evolution toward GPT-5 and responsible, reasoning-centric agents.

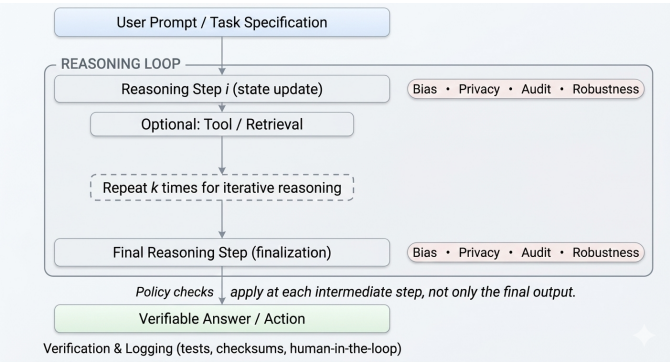


Figure 4: R^2A^2 agent. Each reasoning step calls tools/retrieval and is guarded by Bias, Privacy, Audit, and Robustness checks; the process yields a verifiable answer/action.

- **Self-reflection:** The agent can iteratively evaluate and revise intermediate steps using internal signals or feedback [85].
- **Reinforcement learning:** RL (including RLHF-style feedback) can optimize multi-step trajectories [45].
- **Self-consistency:** Sampling multiple reasoning paths and selecting the most consistent outcome can improve robustness [89].
- **Reasoning reward model:** A reward model can score intermediate or final inferences using internal metrics and/or external feedback, enabling optimization via RL or imitation learning [90].
- **Training pipelines:** Multi-phase fine-tuning, RLHF, and curricula can cultivate robust reasoning behavior [91], and may incorporate fairness objectives and annotated traces to promote safe and valid decision patterns [92].

Together, these components support traceable reasoning with tool use and governance hooks: memory provides grounding, search explores alternatives, and policy/audit mechanisms enable step-level evaluation and oversight.

2.5. Architecture of R^2A^2

To operationalize R^2A^2, we describe a modular architecture with four layers. Figure 5 summarizes the corresponding layered reference architecture.

- **Foundation Layer:** At the foundation is the Core LLM, including tokenization and embeddings for input representation, LayerNorm/residual connections for stable optimization, the transformer stack as the main computation engine, and the pre-training data that shapes semantic and factual knowledge.
- **Reasoning-capable LLM Layer:** This layer adds structured reasoning components. A scratchpad supports CoT/ToT-style step-by-step problem solving, tree search (e.g., MCTS) enables exploration over multiple reasoning paths, self-consistency aggregates candidates for reliability, and (optionally) a reasoning reward model scores intermediate outputs using internal or external feedback.
- *Agentic LLM Layer:* This layer orchestrates task execution. A planner/executor manages goal decomposition and action sequencing, memory supports context persistence, a reflection loop enables iterative self-correction, and a tool interface connects to external services (e.g., databases, calculators, browsers) for tool-augmented actions.
- *Governance & Oversight Layer:* The top layer supports accountability and oversight. A policy engine encodes constraints and boundaries, an audit logger records reasoning steps and tool interactions for *post hoc* analysis, safety/bias checks can be applied at run-

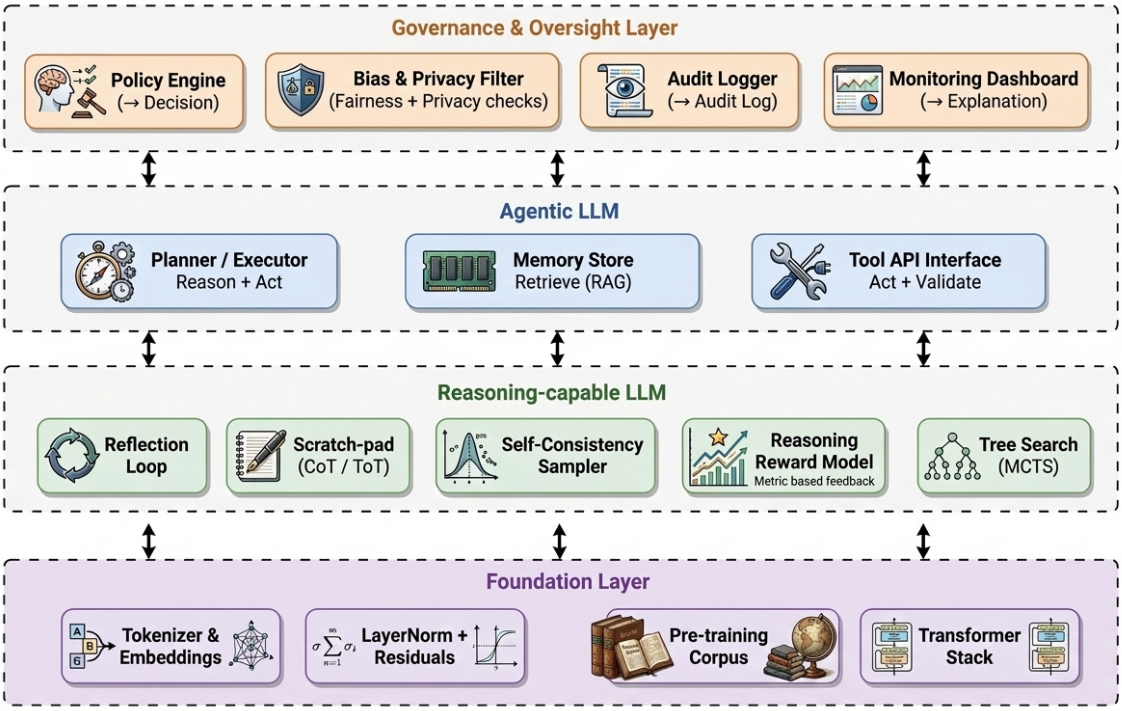


Figure 5: **R^2A^2 reference architecture.** The *Governance & Oversight* layer supports policy enforcement, bias/privacy checks, auditing, and monitoring (e.g., policy engine, bias & privacy filter, audit logger, monitoring dashboard). The *Agent* layer operationalizes a *retrieve–reason–act–validate* loop via planning/execution, memory (e.g., RAG), and a tool interface. The *Reasoning* layer captures explicit inference strategies (e.g., reflection, scratchpad reasoning, self-consistency, reward modeling, and tree search), while the *Foundation* layer represents core LLM components (tokenization/embeddings, transformer stack, and pre-training data).

time, and monitoring supports observability for human supervision.

- *Execution Pipeline: From Ingest to Reason to Act:* Figure 6 illustrates the R^2A^2 execution pipeline, which integrates a functional flow: **Ingest**, **Reason**, and **Act**, with a theoretical hierarchy of information fusion. These three phases align with cross-cutting pathways, which are: GOVERNANCE, CONTROL & EXECUTION, and DATA, to ensure that agentic reasoning remains both performant and responsible.

We ground R^2A^2 in this tradition by mapping classical fusion concepts onto the agentic reasoning pipeline. Following related works, we identify three fusion levels operating within an R^2A^2 agent: (1) **Data-level fusion (Level 0-1).** Operates primarily within the **Data** pathway during **Ingest** and **Act**. Tokenization and embeddings encode text into latents, while RAG modules and tool APIs inject raw evidence such as documents and database records, into the agent's context. (2) **Feature-level fusion (Level 2).** Resides within the CONTROL & EXECUTION and **Data** pathways during the **Reason** phase. Here, the LLM attention mechanism and scratchpad reasoning (utilizing CoT/ToT structures) combine heterogeneous features, including textual semantics and numerical tool outputs, into unified latent representations. (3) **Decision-level fusion (Level 3).** Governed by the GOVERNANCE pathway, where the agent's policy engine aggregates various verdicts, including correctness signals, safety checks, and fairness assessments; into a final validated action or answer. Methods such as self-consistency sampling and majority voting serve as concrete instantiations of this level. Because conflicting evidence is unavoidable in multi-source fusion, the R^2A^2 pipeline includes a dedicated **validate** step within the **Act** phase to serve as a conflict-resolution layer. This ensures that tool outputs or retrieved text do not violate safety constraints or contradict parametric knowledge. All modules report to a **centralized monitoring bus**, which provides real-time oversight and ensures that the fusion process is fully auditable and governance-ready.

3. Progress in Responsible Agentic AI Systems

This section first reviews backbone LLMs for responsible agents. It also discusses agentic web browsers that integrate autonomous AI to complete web-based tasks. Finally, we discuss benchmark datasets and metrics to evaluate the Agentic AI workflows.

3.1. Backbone LLMs for Responsible Agents

Recent backbone LLMs employed in responsible reasoning agents differ along several dimensions relevant to this

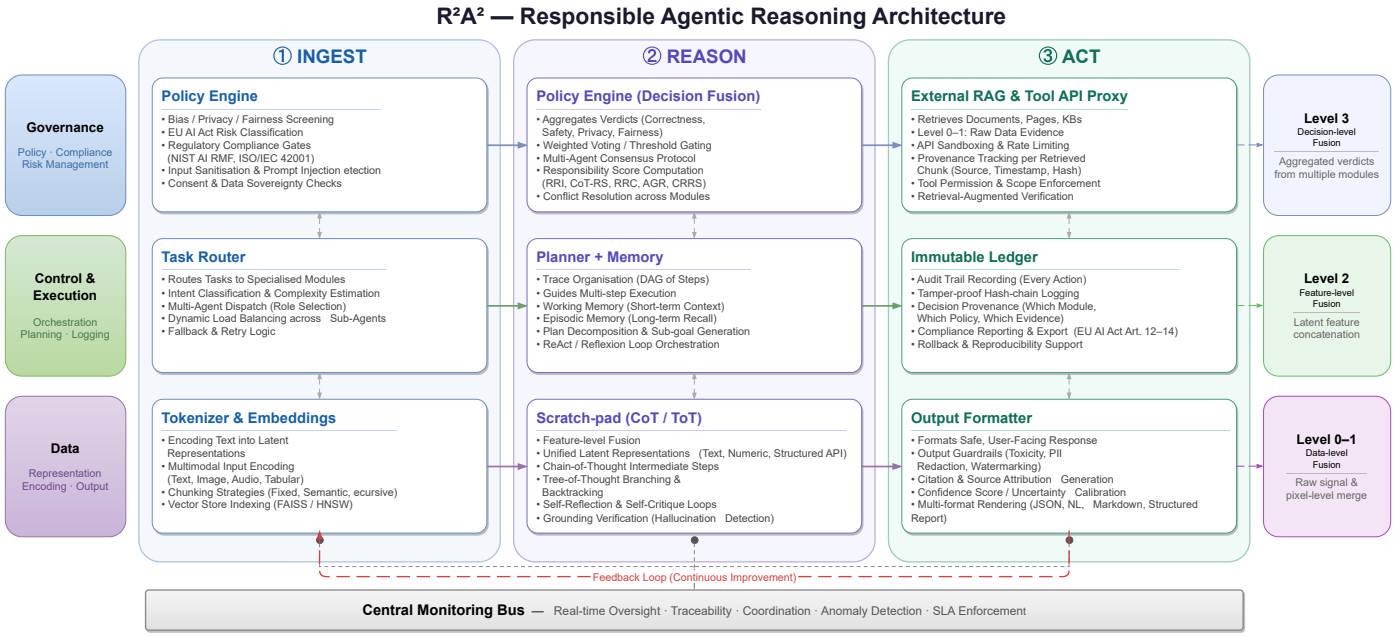


Figure 6: Functional pipeline of a Responsible Reasoning AI Agent (R^2A^2) showing *Ingest–Reason–Act* stages intersected by governance, control/execution, and data pathways. A centralized monitoring bus supports traceability and coordination.

study: reasoning interface (standard, extended, or hybrid thinking), modality support, deployment openness (API-only versus open-weight), and the extent of documented safety and governance mechanisms. Across 2024–2026 the ecosystem evolved from strong proprietary long-context models toward hybrid reasoning systems and open-weight alternatives that increasingly support tool use, structured reasoning traces, and more explicit deployment controls. Table 3 provides a comparison of the backbone models referenced throughout this study.

Four trends emerge from this landscape as shown in Table 3. First, reasoning modes have shifted from fixed long-context inference in 2024 toward controllable reasoning budgets, such as switchable thinking depths (Granite 3.2/3.3, Qwen3), adaptive effort allocation (Claude Opus 4.6, GPT-5.4), and RL-trained trace generation (DeepSeek-R1, QwQ-32B) that gives developers explicit levers over the trade-off among compute cost, latency, and trace transparency that is central to responsible agentic design. Second, multimodal support remains concentrated on the proprietary side (Gemini, GPT-5/5.4, Claude), with Llama 4 as the principal open-weight exception; most auditable, self-hosted options are still text-only, which constrains reproducible research on responsible multimodal agents. Third, the split between open-weight and API-based models (about 6:9) shows a clear trade-off: open-weight models are more transparent, offering access to reasoning and data details, while proprietary models tend to have stronger and more structured safety evaluations. Fourth, and consequently, explicit safety and governance documentation is unevenly distributed: only a minority of entries foreground structured safety mechanisms, while most open-weight releases emphasise capability advances (RL reasoning, long context) without commensurate governance artefacts.

3.2. Agentic Web Browsers

We use the term *agentic web* to denote an internet ecosystem in which autonomous, LLM-driven agents plan, coordinate, and execute goal-directed tasks across websites [143]. Agentic web browsers integrate planning, tool use, and memory to carry out multi-step interactions (plan → navigate → act → verify) under human oversight, making them a natural testbed for R^2A^2: each stage surfaces concerns that map onto core responsibility dimensions including transparency (stepwise action traces), privacy (client-side or sandboxed execution), auditability (immutable action logs), bias mitigation (human-in-the-loop checkpoints), and robustness (least-privilege API design). We categorise recent systems as commercial, research, and open-source platforms; Appendix Figure A1 provides a visual taxonomy and Table 4 summarises representative systems with their responsible-design features and evaluation status.

Four patterns emerge from this landscape. First, *oversight fidelity varies widely*: implementations range from lightweight confirmation prompts (Perplexity Comet, Meteor) through structured authentication takeover (OpenAI Operator, ChatGPT agent) to policy-grounded action filters (Claude in Chrome's Constitutional AI guardrails, Gemini / Project Mariner's AI Principles filtering), however, no current benchmark distinguishes among these levels. Second, *sandboxing is the dominant robustness lever*: virtualised or container-isolated environments (SWE-Agent, CUA, Amazon Q Developer) exhibit qualitatively different failure modes from live-session execution (Opera, Dia), where a single hallucinated action can reach production credentials. Third, *evaluation maturity lags deployment*:

Table 3: Comparison of representative backbone models referenced in this study. **Modality abbreviations:** T = Text, V = Vision, I = Image, A = Audio, Vi = Video, C = Code, Ag = Agentic, CU = Computer-use.

Model	Year	Mod.	Reasoning Mode	Access	Comparative Note
Claude 3.5 Sonnet	2024	T, V	Standard long-context	API / hosted	Strong instruction following with Constitutional AI-style safety posture.
DeepSeek-R1	2025	T	Extended (explicit traces)	Open-weight	MoE architecture with exposed thinking traces; suitable for self-hosted auditing.
Claude 3.7 Sonnet	2025	T, C	Hybrid / extended thinking	API / hosted	Switchable extended-thinking with system-card-style safety evaluation.
Grok 3	2025	T	Extended + search	API / hosted	Frontier model emphasising deep search and multi-step problem solving.
Granite 3.2/3.3	2025	T, C	Controllable thinking	Open-weight	Enterprise-oriented with data-provenance emphasis and governance-friendly documentation.
QwQ-32B	2025	T	RL-trained extended	Open-weight	Dense reasoning model with verifier-based RL and long context.
Gemini 2.5 Pro	2025	T, I, A, Vi	Extended thinking	API / hosted	Natively multimodal with large context and strong tool-use support.
OpenAI o3 / o4-mini	2025	T, V	Multi-step reasoning	API / hosted	Optimised for tool-augmented workflows and deliberative safety alignment.
Llama 4	2025	T, I	Hybrid multimodal	Open-weight	Large-context open-weight family with broad deployment flexibility.
Qwen3	2025	T	Hybrid thinking / non-thinking	Open-weight	Controllable reasoning budget with strong agentic tool-use support.
Skywork-OR1	2025	T	RL-based reasoning	Open-weight	Lightweight series designed for reproducible RL-based reasoning research.
GPT-5	2025	T, V	Unified adaptive	API / hosted	Configurable reasoning effort with strengthened safety and factuality controls.
Gemini 3 Pro	2025	T, I, A, Vi	Deep Think capable	API / hosted	Flagship multimodal model with formalised reasoning mode.
Claude Opus 4.6	2026	T, C, Ag	Adaptive effort-based	API / hosted	High-capability reasoning with strong safety controls and low over-refusal emphasis.
GPT-5.4	2026	T, V, CU	Adaptive + tool search	API / hosted	Native computer use, long context, and improved factuality for agentic workflows.

roughly half the entries lack any standardised benchmark result, and the available benchmarks (WebVoyager, OS-World, SWE-Bench, WebArena, BEARCUBS) differ in task ontology, privilege scope, and success criteria, making cross-system responsibility comparisons unreliable. Fourth, *transparency is stratifying* from passive action traces toward active safety monitoring, e.g., most notably OpenAI CUA's chain-of-thought deception detection and cyber-risk classification, signalling a shift whose generalisability will be an important axis for future R^2A^2 evaluation.

3.3. Benchmark Datasets and Evaluation Metrics

To rigorously evaluate the reasoning and problem-solving capabilities of LLM-based agentic systems, researchers employ a diverse suite of benchmarks spanning mathematical reasoning, coding, abstract generalization, multimodal understanding, commonsense inference, and interactive tool use. Table 5 summarizes representative benchmarks commonly used to assess LLM-based agents.

Mathematical reasoning benchmarks. Datasets such as GSM8K [109], AIME [110], MATH [111], IMO-style problems [144], and FrontierMath [113] evaluate reasoning ability across difficulty levels, from grade-school arithmetic to competition- and research-level mathematics. Performance is typically measured by exact answer accuracy. GSM8K contains thousands of multi-step arithmetic word problems designed to test compositional reasoning. The MATH dataset includes competition-style problems across algebra, geometry, number theory, and combinatorics. Olympiad-style problems such as AIME and IMO remain challenging under standard prompting, often requiring deeper symbolic reasoning. FrontierMath introduces expert-curated problems spanning advanced domains (e.g., number theory, algebraic geometry), where even strong models solve only a small fraction, highlighting the gap between LLM performance and human experts.

Knowledge and commonsense reasoning. MMLU (Massive Multitask Language Understanding) [145] and GPQA [114] assess broad factual knowledge and domain-specific reasoning across multiple disciplines. MMLU spans dozens of subjects using multiple-choice questions, with accuracy as the primary metric. GPQA targets graduate-level scientific reasoning and is designed to reduce simple memorization or lookup strategies. CommonsenseQA [116] evaluates everyday inferential reasoning beyond explicitly stated facts. Standard evaluation relies on accuracy over MCQs.

Abstract and general reasoning. ARC-AGI (Abstraction and Reasoning Corpus) [146] measures abstract reasoning and generalization by requiring agents to infer latent rules from a small number of examples and apply them to novel inputs. Performance is measured by success rate on previously unseen tasks. Recent updates [139] introduce more fine-grained challenges aimed at quantifying progress toward human-like general problem solving.

Coding and software engineering. HumanEval [121] evaluates code generation by requiring models to produce Python functions that pass hidden unit tests. The standard metric, pass@k, measures the probability that at least one of k generated samples passes all tests, with pass@1 reflecting first-attempt correctness. SWE-Bench [122] extends evaluation to real-world software engineering by including thousands of GitHub issue-resolution tasks. Metrics typically include functional correctness and issue-level success rate. LiveCodeBench [123] curates competitive programming tasks released after model training cutoffs to reduce data leakage; evaluation uses automated judges to compute the fraction of correctly solved tasks and may incorporate iterative self-repair settings.

Multimodal benchmarks. As agents increasingly integrate perception and action, evaluation has expanded to multimodal reasoning tasks. MMMU [124] contains over ten thousand questions requiring joint vision–language reasoning across multiple academic domains, with accuracy as the primary metric. GAIA [125] evaluates tool-augmented, multi-step reasoning in realistic task settings that may involve textual, visual, or structured inputs. Performance is measured by task completion rate or correctness under autonomous tool-use conditions.

Interactive and tool-use benchmarks. Beyond static question answering and code generation, interactive benchmarks evaluate agents operating within dynamic environments that require sequential decision-making, real-time adaptation, and multi-step tool orchestration. Web-navigation benchmarks such as WebShop [147], WebArena [127], and Mind2Web [148] assess whether agents can achieve user-specified goals (e.g., purchasing a product, filling a form).

Evaluation in these settings goes beyond correctness to encompass task completion rate, step efficiency, and constraint violations, capturing both whether the agent succeeded and how efficiently and safely it did so. Environment-based benchmarks such as ALFWorld [149] and BabyAI [150] complement web-focused tasks by evaluating grounded instruction following in simulated household and grid-world environments, where agents must map natural-language directives to executable action sequences over multiple turns. AgentBench-style evaluations [126] further extend this paradigm to diverse tool-use scenarios spanning database querying, API composition, operating-system interaction, and lateral-thinking puzzles, measuring success rate over multi-step action sequences that demand both planning and error recovery. Collectively, these benchmarks shift evaluation from single-inference accuracy toward trajectory-level assessment, making them particularly relevant to R^2A^2, where each intermediate action constitutes a potential site for responsibility checks including transparency of intent, privacy of accessed data, and auditability of the full action trace.

Agent-level metrics. In addition to task-level metrics (e.g., exact match, pass@k, BLEU), LLM-based agents are increasingly evaluated using process- and trajectory-level measures. These include task completion rate, number of steps to success, tool-call utility (fraction of tool invocations that contribute meaningful evidence), and recovery from intermediate failures or dead ends [151]. Efficiency is typically measured via wall-clock time, total tokens consumed, and compute cost per successful episode. Some evaluations further assess plan–action consistency and evidential support, examining whether the agent's final answer is justified by intermediate reasoning traces, tool outputs, or retrieved documents [25].

Safety, robustness, and fairness. For deployment-oriented evaluation, benchmark design increasingly incorporates safety and robustness criteria. Common metrics include harmful or policy-violating output rate, robustness to jailbreak or adversarial prompts, and hallucination or groundedness rate when interacting with external tools or web sources [22]. Fairness-aware evaluations measure group-wise performance gaps and disparate error patterns across demographic attributes, while robustness analyses examine performance under distribution shift, noisy tool outputs, or incomplete information. Together, these dimensions complement task success metrics and provide a more realistic assessment of agent readiness in high-stakes environments.

Despite this breadth, current benchmarks remain fragmented: few jointly evaluate reasoning quality, tool-use behavior, and responsibility properties such as fairness or privacy, limiting their applicability for R^2A^2 style evaluation. An analysis of the reported results, as shown in Table 5, reveal a substantial gap between current agents and human performance, for example, on GAIA, even top-performing systems on GAIA achieve at most ∼65% success compared to 92% for human participants, and on ARC-AGI-2 the best model (Gemini 3 Deep Think, 84.6%) still falls short of human-level generalization, while SWE-Bench Verified scores have surged past 80% (Claude Opus 4.5/4.6, GPT-5.2), suggesting that software-engineering tasks may be approaching practical automation thresholds, while specialized frameworks (e.g., h2oGPT-e, Refact.ai) show promising improvements in domain-specific settings. Overall, there is a need for planning, multi-step reasoning, and robust tool use, reinforcing the importance of continued progress in agent architecture design and evaluation.

Table 4: Representative responsible–reasoning web/agent systems (2024–2026)

Agent System	Type & Domain	Responsible Design Features	Evaluations / Status
Opera Browser Operator [93]	AI-integrated browser (client-side)	On-device execution; stepwise action traces; human override	Feature preview (Mar 2025); public demos of end-to-end shopping
Dia (The Browser Company) [94]	AI-first browser	Command/URL bar; opt-in features; evolving privacy controls	Private beta (Jun 2025); no public benchmark yet
Perplexity Comet [95]	AI-native browser	Assistant sidebar; confirmations before consequential actions	Launch (Jul 9, 2025); hands-on reports note tab-aware actions
Meteor [96]	AI-native agentic browser	Task-describing command bar; explicit approval for actions that modify state	YC-backed startup (2025); early user reports highlight autonomous tab flows
Fellou [97]	Spatial agentic AI browser	Full workflow transparency with editable step-by-step plans; user can approve or intervene at any step; agentic memory with local context	Launched 2025; hybrid natural-language + JavaScript agent scripting
OpenAI Operator [98]	Web agent (virtual browser)	Takeover for auth/payments; explicit confirmations; narrated steps	Research preview (Jan 2025); later unified into ChatGPT agent
OpenAI ChatGPT agent [63]	Web+software agent (inside ChatGPT)	User confirmation; Watch mode; secure takeover	Public rollout (Jul 2025); results vary by benchmark
ChatGPT Atlas [99]	Standalone AI browser (macOS)	Agent mode with visible action plans; user supervision for navigation and transactions	Initial release (Oct 2025); positioned as AI-first alternative to traditional browsers
OpenAI CUA (GPT-5.4) [63]	Computer-use agent (API)	Screenshot + keyboard/mouse interaction; high cyber-risk classification; CoT monitoring for deception detection; trusted access controls	Released Mar 2026; **75.0%** on OSWorld-Verified (human: 72.4%); 67.3% on WebArena-Verified
SWE-Agent [100]	Autonomous coding agent	Docker sandbox; verify&edit loop; limited network	**12.5%** pass@1 on SWE-Bench (original); guarded execution
Biotic Browser [101]	Research browser prototype	Streaming LLM co-pilot; persistent session context; user-in-the-loop control	CHI-style research prototype; qualitative user studies, no standardized benchmark
LiteWebAgent [102]	Open-source web-agent suite	CDP/remote browser; planning and memory modules	NAACL 2025 demo; Chrome extension + remote driver
CowPilot [103]	Human-agent collaborative web agent	Framework for autonomous and human-agent collaborative navigation; user can intervene at any step	arXiv 2025; designed for shared human–AI control
Gemini 2.0 / Project Mariner [67]	Web agent (virtual browser)	Multimodal page understanding; user confirmation for consequential actions; Google AI Principles filtering	Research preview (Dec 2024); 83.5% on WebVoyager
Browser Use [104]	Open-source browser agent framework	Playwright-based agent control; structured action logging; extensible with any LLM backend	**89.1%** on WebVoyager (586 tasks); leading open-source web-agent framework
GitHub Copilot Coding Agent [81]	Enterprise coding agent	Draft-PR workflow; branch protections; audit logs	Automates low/medium repo tasks via Actions
Amazon Q Developer [105]	Agentic IDE assistant	Read/write/run in IDE; vulnerability scan; IAM-scoped permissions	Multi-step task automation in VS Code/JetBrains/VS
AWS Strands Agents [106]	Open-source agents SDK	Model-driven orchestration; observability	v1.0 release (Jul 2025); production-oriented toolkit
Mistral Agents API [107]	Enterprise agent platform	Developer-defined tools; on-prem options; open-weights	API for planning/tool use; docs and examples
Claude in Chrome [68]	AI browser extension (Chrome)	Constitutional AI guardrails at browser-action level; user oversight; Anthropic safety policies applied to navigation and form actions	Released Aug 2025; pairs with Claude Code for agentic browser control; no standardized web benchmark yet
Microsoft AutoGen 0.4 [31]	Open-source multi-agent framework	Modular agent orchestration; human-in-the-loop hooks; configurable guardrails	v0.4 release (2025); supports multi-agent web and coding workflows
Google Chrome Auto Browse [67]	AI feature in Chrome browser	Autonomous task completion via Gemini 3 side panel; Premium-subscriber gated; user confirmation for state-modifying actions	Launched Jan 2026; integrated into Chrome for Premium subscribers
Microsoft Edge Copilot Mode [108]	AI-integrated browser (Edge)	Multi-tab context reasoning; Copilot Actions for autonomous form-filling and task execution; toggleable privacy and memory controls	Launched Oct 2025; free, built into Windows; no standardized web-agent benchmark yet

Table 5: Benchmark datasets for evaluating AI reasoning, problem-solving, and agentic capabilities, with abbreviated evaluation metrics.

Dataset	Purpose	Domain	Key characteristics	Typical metrics
GSM8K (Grade School Math 8K) [109]	Mathematical reasoning	Grade-school math	8,000 word problems testing arithmetic and logical reasoning	Acc (EM)
AIME (American Invitational Mathematics Examination) [110]	Advanced mathematical reasoning	High-school math	Challenging problems from the AIME contest; yearly sets (e.g., AIME24, AIME25) used as AI benchmarks requiring multi-step symbolic reasoning	Acc
MATH [111]	Complex problem solving	HS/college math	Advanced, multi-step problems requiring deeper reasoning	Acc
International Mathematical Olympiad (IMO) [112]	Advanced reasoning	Olympiad math	High-difficulty problems from past IMO contests	Score, SR
FrontierMath [113]	Advanced mathematical reasoning	Research-level math	Expert-curated problems spanning number theory, algebraic geometry, and other advanced domains; even strong models solve only a small fraction	Acc
GPQA (General-Purpose Question Answering) [114]	Broad question answering	General knowledge	Graduate-level science questions, Google-proof	Acc
MMLU (Massive Multitask Language Understanding) [115]	Multitask knowledge evaluation	Academic / professional	57 tasks spanning STEM, humanities, and professional subjects	Acc
CommonsenseQA [116]	Commonsense reasoning	General knowledge	Multiple-choice questions testing commonsense understanding and inference	Acc
BIG-Bench [117]	Multitask reasoning	Language and reasoning	Diverse tasks testing analogy, causality, and commonsense reasoning	Acc, F1
Humanity's Last Exam (HLE) [118]	Frontier knowledge evaluation	Multi-domain expert	Expert-sourced questions designed to resist memorization; intended as a ceiling benchmark for frontier models	Acc
GDPval [63]	Knowledge-work evaluation	Professional / enterprise	Measures model performance on economically valuable tasks across 44 occupations (law, logistics, engineering, sales, etc.)	Acc
ARC-AGI (Abstraction & Reasoning Corpus) [119]	Abstract reasoning	General intelligence	Visual and logical tasks to test abstraction and generalisation	SR
Codeforces Elo Rating [120]	Competitive programming	Coding	Elo-based metric from Codeforces for algorithmic problem-solving skill	SR, Elo
HumanEval [121]	Code generation	Programming	164 coding problems with unit tests for functional correctness	P@1, P@k
SWE-Bench [122]	Software engineering	Coding	Real-world GitHub issues for debugging and implementation	TCR, FC
LiveCodeBench [123]	Dynamic coding evaluation	Programming	Continuously updated problems for real-time performance measurement	PR, SR
MMMU (Massive Multimodal Understanding) [124]	Multimodal reasoning	Multimodal (text, images)	Complex tasks combining text and visual reasoning across STEM and humanities	Acc
HealthBench [63]	Medical / health QA	Healthcare	Realistic medical scenarios with physician-defined evaluation criteria; Hard subset targets complex clinical reasoning	Acc
GAIA (General AI Assistant) [125]	Agent-based reasoning	Real-world tasks	Multi-step reasoning, tool use, and cross-modal understanding	TCR, TUA, Steps
AgentBench [126]	Multi-environment agent evaluation	Real-world tasks	Eight diverse environments (web browsing, household, DB, etc.) testing tool use and multi-step decision-making	SR, TCR
WebArena [127]	Web navigation agent evaluation	Web tasks	Realistic self-hosted web environments (shopping, forum, CMS, map); tasks require multi-step interaction under constraints	TCR, SR
WebVoyager [128]	End-to-end web agent evaluation	Web tasks	Real-website navigation tasks requiring visual understanding and multi-step action sequences	SR
BrowseComp [129]	Agentic web search evaluation	Web retrieval	Tasks requiring persistent, multi-step web browsing to locate hard-to-find information	Acc
OSWorld [130]	Desktop computer-use evaluation	OS-level tasks	Real desktop environments (Ubuntu, Windows, macOS); agents navigate via screenshots and keyboard/mouse actions	SR
Terminal-Bench [131]	Command-line agent evaluation	Software engineering	Multi-step terminal tasks requiring file manipulation, debugging, and system administration	SR

Note: Acc = Accuracy, EM = Exact Match, P@1/P@k = Pass at $1/k$ attempts, SR = Solve Rate, PR = Pass Rate, TCR = Task Completion Rate, TUA = Tool-Use Accuracy, FC = Functional Correctness, F1 = F1 Score, Score = IMO points (0–42).

Table 6: Performance of LLM-based agents on public agentic benchmarks (2024-2026). Values are as reported in source publications; metrics such as tool-use accuracy or step count are omitted here as they are infrequently reported.

Benchmark	Agent/Model	Success Rate (%)	Notes
GAIA [125]	Human (degree holders)	92	Human average performance.
	GPT-4 (with plugins)	15	With web browsing/tools; struggled on GAIA.
	GPT-4 (no plugins)	≈7	GPT-4 Turbo baseline agent scored under 7%.
	Autogen Multi-Agent [31]	40	Multi-agent using GPT-4 with tool-calls.
	h2oGPT-e Agent [132]	∼65	Latest open-source agent (Feb 2025), ∼27% below human.
	GPT-5 (agent mode) [133]	–	Significantly improved over GPT-4 with plugins; exact GAIA figure not yet standardized.
AgentBench [126]	GPT-4 (OpenAI2025)	∼45	Best on 7/8 tasks (e.g., 78% on Household).
	GPT-3.5-Turbo	∼18	Much lower than GPT-4; e.g., 21% on web browsing vs GPT-4's 55%.
	LLaMA2-70B [134, 135]	20.4	Strongest open model; 20.4% success.
MLAgentBench [136]	Claude v3 Opus	37.5	Highest average success across 13 ML experiment tasks.
	GPT-4 (research agent) [137]	–	90% on old datasets but 0–10% on novel tasks.
	Gemini-Pro, GPT-4 Turbo	< 37	Trailed Claude v3 on average.
ARC-AGI [138]	Human	97–99	Humans solved ∼98% of private ARC tasks.
	OpenAI O3 (2025) [63]	75.7	Fine-tuned O3 model; 87.5% with 172× compute.
	GPT-4 (GPT-4o)	9	Solved only 5–9% of ARC puzzles.
	Claude 3.5 [68]	21	Sonnet solved ∼21%.
	O1 preview	21	GPT-based O1 scored ∼21%.
	Gemini 1.5	∼8	Scored ∼8% on ARC public eval.
	ARC Prize SOTA (2024)	55.5	Program-synthesis + test-time training approach.
ARC-AGI-2 [139]	Gemini 3 Deep Think [67]	84.6	Highest published score (Dec 2025).
	Claude Opus 4.6 [68]	68.8	83% improvement over Opus 4.5 (Feb 2026).
	GPT-5.2 Pro [63]	54.2	OpenAI Pro-tier reasoning (Dec 2025).
	Claude Opus 4.5 [68]	37.6	Nov 2025 baseline.
MobileAgent Bench [140]	M3A Agent [140]	64.0	Best single-app success rate.
	T3A (Vision+LLM) [140]	48.7	Tencent vision-enabled agent.
	MobileAgent V2[3]	43.3	Second-best open-source mobile agent.
	SeeAct	39.3	Visual UI agent.
	AppAgent (GPT-4)	34.0	Baseline GPT-4 mobile agent.
	Fine-tuned RL Agents	1–5	Learned policies struggled (1–5%).
WebArena [127]	Human reference	78	Human success on WebArena.
	IBM CUGA[4]	61.7	Hierarchical web agent; planner + subagents.
	OpenAI Operator [63]	58.1	GPT-4V ChatGPT Agent mode.
	Zeta Jace.AI[5]	57.1	Startup autonomous web agent.
	ScribeAgent + GPT-4[6]	53.0	Hybrid open agent.
	AgentSymbiotic[7]	52.1	Multi-agent LLM symbiosis.
	Occam	48.5	Prompt-engineered GPT-4 agent.
	Learn-by-Interact	48.0	Google RL-tuned agent.
SWE-Bench Lite [122]	Refact.ai Agent [141]	59.7	Top of SWE-Bench Lite leaderboard.[8]
	Globant Code Fixer	48.3	Multi-agent code fixer.
	SWE Lite Baseline [122]	43	Earlier SOTA on Lite benchmark.
	GPT-4 (Code)	∼20	GPT-4 without loop agent.
SWE-Bench Verified	Claude Opus 4.5 [68]	80.9	Highest at release (Nov 2025).
	Claude Opus 4.6 [68]	80.8	Feb 2026; with agent teams.
	GPT-5.2 [63]	80.0	Dec 2025; closes gap with Claude.
	GPT-5 [63]	74.9	Aug 2025 baseline.
	DeepSeek V3.2 [142]	–	GPT-5-class (reported); exact figure not standardized.
BrowseComp[9]	Claude Opus 4.6 [68]	84.0	Agentic web search (Feb 2026).
	Claude Opus 4.5 [68]	67.8	Nov 2025.

Table 7: Multi-dimensional comparison of representative works across R^2A^2 responsibility dimensions. ✓ = primarily addressed, △ = partially addressed, ✗ = not addressed. RS = Reasoning (CoT/ToT/GoT), TU = Tool Use & RAG, FB = Fairness/Bias, PV = Privacy, TE = Transparency/Explainability, AG = Auditability/Governance, TL = Trace-level Evaluation, SR = Safety/Robustness, OS = Open Artifacts*. Works are grouped by thematic category; N=115 representative papers are selected from our corpus of reviewed works.

Work	Type	Year	RS	TU	FB	PV	TE	AG	TL	SR	OS*
A. Reasoning Methods											
CoT Prompting [9]	Method	2022	✓	✗	✗	✗	✓	✗	△	✗	✗
ReAct [10]	Method	2023	✓	✓	✗	✗	✓	✗	△	✗	✓
Tree-of-Thought [11]	Method	2023	✓	✗	✗	✗	✓	✗	△	✗	✓
Graph-of-Thought [29]	Method	2024	✓	✗	✗	✗	✓	✗	△	✗	✓
Self-Consistency [89]	Method	2022	✓	✗	✗	✗	△	✗	△	✓	✗
Logic-of-Thought [152]	Method	2024	✓	✗	✗	✗	✓	✗	△	✗	✗
Self-Refine [153]	Method	2023	✓	✗	✗	✗	✓	✗	✓	△	✓
Reflexion [54]	Method	2023	✓	△	✗	✗	✓	✗	✓	△	✓
LLaMA-Berry [154]	Method	2024	✓	✗	✗	✗	△	✗	✓	△	✓
Soft Thinking [155]	Method	2025	✓	✗	✗	✗	△	✗	△	✗	✗
MRKL Systems [156]	Method	2022	△	✓	✗	✗	△	✗	✗	✗	✗
B. Backbone LLMs for Responsible Agents											
GPT-4 / GPT-4o [63]	Model	2024	✓	✓	△	△	△	△	△	✓	✗
GPT-5† [63]	Model	2025	✓	✓	△	✓	△	△	△	✓	✗
OpenAI o3 / o4-mini [63]	Model	2025	✓	✓	△	△	✓	△	✓	✓	✗
GPT-5.1 Thinking† [63]	Model	2025	✓	✓	△	△	✓	△	△	✓	✗
GPT-5.2† [63]	Model	2025	✓	✓	△	✓	✓	✓	△	✓	✗
gpt-oss-120B† [63]	Model	2025	✓	✓	△	△	✓	✓	△	△	✓
Claude 3.5 Sonnet [68]	Model	2024	✓	✓	△	✓	△	△	△	✓	✗
Claude 3.7 Sonnet [68]	Model	2025	✓	✓	△	✓	✓	△	✓	✓	✗
Claude Sonnet 4 [68]	Model	2025	✓	✓	△	✓	✓	△	✓	✓	✗
Claude Opus 4.5 [68]	Model	2025	✓	✓	✓	✓	✓	✓	✓	✓	✗
Claude Opus 4.6† [68]	Model	2026	✓	✓	✓	✓	✓	✓	✓	✓	✗
DeepSeek R1 [157]	Model	2025	✓	△	△	△	✓	△	✓	△	✓
QwQ-32B [158]	Model	2025	✓	△	△	△	△	△	△	△	✓
Gemini 2.5 Pro [67]	Model	2025	✓	✓	△	△	△	△	△	✓	✗
Gemini 3 Pro† [67]	Model	2025	✓	✓	△	△	✓	✓	✓	✓	✗
Gemini Deep Think† [67]	Model	2025	✓	△	✗	△	✓	△	✓	△	✗
Skywork OR1 [159]	Model	2025	✓	✗	✗	✗	△	△	△	✗	✓
DeepSeek V3.2† [142]	Model	2025	✓	✓	△	△	△	△	△	△	✓
Llama 4 [70]	Model	2025	✓	✓	△	△	△	△	△	△	✓
Grok 3 [71]	Model	2025	✓	✓	△	△	△	△	△	✓	✗
IBM Granite 3.2/3.3 [72]	Model	2025	✓	✓	△	△	✓	✓	△	✓	✓
ERNIE 4.5† [73]	Model	2025	✓	✓	△	△	△	△	△	△	✗
Mistral Medium 3 [74]	Model	2025	✓	✓	△	△	△	△	△	△	△
C. Agentic Frameworks & Architectures											
LangChain [30]	Framework	2023	△	✓	✗	✗	△	△	△	△	✓
AutoGPT [62]	Framework	2023	△	✓	✗	✗	△	✗	✗	△	✓
AutoGen v0.4 [31]	Framework	2025	✓	✓	✗	△	△	△	△	△	✓
MetaGPT [160]	Framework	2023	✓	✓	✗	✗	△	△	△	✗	✓
CrewAI [161]	Framework	2023	△	✓	✗	✗	△	△	✗	△	✓
HuggingGPT [162]	Framework	2023	△	✓	✗	✗	△	✗	✗	✗	✓
Generative Agents [163]	Framework	2023	△	△	✗	△	✓	✗	△	✗	✓
Voyager [164]	Framework	2023	✓	✓	✗	✗	△	✗	△	△	✓
CAMEL [165]	Framework	2023	△	△	✗	✗	△	✗	✗	✗	✓
MemGPT [49]	Framework	2023	△	✓	✗	△	△	✗	✗	✗	✓
ExpeL [46]	Framework	2024	✓	△	✗	✗	✓	✗	✓	△	✓
VisualChatGPT [166]	Framework	2023	△	✓	✗	✗	△	✗	✗	✗	✓
Amazon Bedrock AgentCore [80]	Platform	2026	△	✓	△	✓	△	✓	△	✓	✗
AWS Strands Agents [106]	Framework	2025	△	✓	✗	△	△	✓	△	△	✓
Mistral Agents API [107]	Platform	2025	△	✓	✗	△	△	△	✗	△	△
D. Tool Use & Retrieval-Augmented Generation											
RAG [6]	Method	2020	△	✓	✗	△	△	✗	✗	△	✓
Toolformer [88]	Method	2023	△	✓	✗	✗	△	✗	✗	✗	✗
Gorilla [167]	Method	2024	△	✓	✗	✗	△	✗	△	△	✓
API-Bank [168]	Bench	2023	△	✓	✗	✗	△	✗	△	✗	✓
E. Agentic Web Browsers											
OpenAI Operator [63]	Agent	2025	✓	✓	✗	✓	✓	△	△	✓	✗
ChatGPT Agent [63]	Agent	2025	✓	✓	✗	✓	✓	△	△	✓	✗
ChatGPT Atlas [99]	Agent	2025	✓	✓	✗	△	✓	△	△	✓	✗
Claude in Chrome [68]	Agent	2025	✓	✓	△	✓	✓	✓	△	✓	✗
Opera Browser Operator [93]	Agent	2025	△	✓	✗	✓	✓	△	△	△	✗
Perplexity Comet [95]	Agent	2025	△	✓	✗	△	✓	△	✗	△	✗
Dia Browser [94]	Agent	2025	△	✓	✗	△	△	✗	✗	△	✗
Gemini 2.0 / Proj. Mariner [67]	Agent	2024	✓	✓	✗	△	△	△	△	✓	✗
SWE-Agent [100]	Agent	2024	✓	✓	✗	✗	△	△	△	✓	✓
LiteWebAgent [102]	Agent	2024	△	✓	✗	✗	△	✗	△	✗	✓

Continued on next page

Table 7 continued from previous page

Work	Type	Year	RS	TU	FB	PV	TE	AG	TL	SR	OS*
CowPilot [103]	Agent	2025	△	✓	✗	△	△	✗	△	△	✓
GitHub Copilot Agent [81]	Agent	2025	△	✓	✗	△	△	✓	△	△	✗
Amazon Q Developer [105]	Agent	2025	△	✓	✗	✓	△	✓	△	✓	✗
Meteor [96]	Agent	2025	△	✓	✗	△	✓	△	△	△	✗
Biotic Browser [101]	Agent	2024	△	✓	✗	△	△	✗	△	△	✓
Orca [169]	Agent	2025	△	✓	✗	△	△	✗	△	△	✓
Fellou [97]	Agent	2025	△	✓	✗	△	△	✗	✗	△	✗
Opera Neon [170]	Agent	2025	△	✓	✗	△	△	✗	✗	△	✗
F. Responsible AI, Safety & Governance											
Constitutional AI [171]	Method	2022	✗	✗	✓	△	✓	△	✗	✓	✗
Fair LLMs Survey [14]	Survey	2024	✗	✗	✓	△	✓	△	✗	△	✗
Responsible LLMs [16]	Survey	2025	△	△	✓	✓	✓	✓	△	✓	✗
Fairness in Agentic AI [92]	Method	2025	△	✗	✓	△	✓	✓	△	△	✗
Policy Frameworks for CoT [84]	Method	2025	✓	✗	△	△	✓	✓	✓	△	✗
Responsible AI Agents [61]	Method	2025	△	△	✓	✓	✓	✓	△	✓	✗
TRiSM for Agentic AI [21]	Survey	2025	△	△	✓	✓	✓	✓	△	✓	✗
Agentic AI Security [22]	Survey	2025	△	△	△	✓	△	✓	△	✓	✗
Multi-Agent Risks [24]	Survey	2025	✗	△	△	△	△	✓	✗	✓	✗
RAI Metrics Catalogue [172]	Framework	2024	✗	✗	✓	✓	✓	✓	△	△	✓
MultiTrust [173]	Bench	2024	✗	✗	✓	✓	✓	△	△	✓	✓
G. Benchmarks & Evaluation Frameworks											
GAIA [125]	Bench	2023	✓	✓	✗	✗	△	✗	△	✗	✓
AgentBench [126]	Bench	2023	✓	✓	✗	✗	△	✗	△	△	✓
SWE-Bench [122]	Bench	2023	△	✓	✗	✗	△	✗	△	✗	✓
ARC-AGI-2 [139]	Bench	2025	✓	✗	✗	✗	△	✗	△	✗	✓
GSM8K [109]	Bench	2021	✓	✗	✗	✗	△	✗	✗	✗	✓
MATH [111]	Bench	2021	✓	✗	✗	✗	△	✗	✗	✗	✓
FrontierMath [113]	Bench	2024	✓	✗	✗	✗	△	✗	✗	✗	△
HELM [174]	Bench	2023	△	△	✓	△	✓	△	△	✓	✓
Holistic Agent Leaderboard [175]	Bench	2025	✓	✓	△	✗	✓	△	✓	△	✓
Trace Evaluation Survey [18]	Survey	2025	✓	△	✗	✗	✓	△	✓	△	✗
WebArena [127]	Bench	2024	△	✓	✗	✗	△	✗	△	△	✓
MMMU [124]	Bench	2024	✓	✗	✗	✗	△	✗	✗	✗	✓
LiveCodeBench [123]	Bench	2024	△	✗	✗	✗	△	✗	△	✗	✓
GPQA [114]	Bench	2024	✓	✗	✗	✗	△	✗	✗	✗	✓
BIG-Bench [176]	Bench	2023	✓	△	△	✗	△	✗	✗	△	✓
WebVoyager [128]	Bench	2024	△	✓	✗	✗	△	✗	△	△	✓
Mind2Web [148]	Bench	2023	△	✓	✗	✗	△	✗	△	△	✓
WebShop [147]	Bench	2022	△	✓	✗	✗	△	✗	△	△	✓
ALFWorld [149]	Bench	2020	✓	✓	✗	✗	△	✗	△	△	✓
MLAgentBench [136]	Bench	2023	✓	✓	✗	✗	△	✗	△	△	✓
MobileAgentBench [177]	Bench	2025	△	✓	✗	△	△	✗	△	△	✓
AssistantBench [178]	Bench	2024	✓	✓	✗	✗	△	✗	△	△	✓
BEARCUBS [179]	Bench	2025	△	✓	✗	✗	△	✗	△	△	✓
H. Surveys on Reasoning & Agentic AI											
LLM Reasoning Survey [2]	Survey	2024	✓	△	△	✗	△	✗	△	△	✗
Agentic LLMs Survey [15]	Survey	2025	✓	✓	△	✗	△	△	△	△	✗
AI Agents vs. Agentic AI [3]	Survey	2025	✓	✓	△	△	△	△	△	△	✗
Agentic Reasoning [4]	Survey	2026	✓	✓	✓	△	✓	△	△	✓	✗
RLM Blueprint [19]	Survey	2025	✓	△	△	✗	△	✗	△	△	✗
100 Days After DeepSeek [20]	Survey	2025	✓	△	✓	△	✓	△	△	✓	✗
Deep Research Agents [26]	Survey	2025	✓	✓	△	✗	△	✗	△	△	✗
LM Agent Cooperation [27]	Survey	2025	△	△	✓	△	△	△	△	✓	✗
Internet of Agents [28]	Survey	2025	✗	△	△	△	△	△	✗	✓	✗
Large Reasoning Models [23]	Survey	2025	✓	△	△	✗	△	✗	△	△	✗
I. This Survey											
R^2A^2 (this work)	**Survey**	**2026**	✓	✓	✓	✓	✓	✓	✓	✓	✓
Summary Statistics (excluding R^2A^2; $N = 115$)											
Primarily addressed (✓)			66	72	14	19	42	19	15	35	57
Partially addressed (△)			43	21	33	44	73	42	76	54	3
Not addressed (✗)			6	22	68	52	0	54	24	26	55
Coverage (✓ + △)			**95%**	**81%**	**41%**	**55%**	**100%**	**53%**	**79%**	**77%**	**52%**

† Assessment based on vendor documentation and system cards rather than independent third-party evaluation; responsibility ratings may shift as independent audits emerge.

* **OS (Open Artifacts)** encompasses open-source code for Methods and Frameworks, open weights for Models, and publicly released data for Benchmarks.

3.4. Qualitative synthesis: strengths and limitations by category

Beyond the quantitative summary, Table 7 reveals systematic trade-offs across method families that merit qualitative discussion.

Reasoning methods (Category A) excel at transparency and trace-level evaluation through inspectable intermediate steps (CoT, ToT, self-refinement), but are nearly silent on fairness, privacy, and governance, a structural gap since responsible reasoning requires trace mechanisms that surface bias and privacy violations alongside debugging. **Backbone LLMs** (Category B) offer the broadest dimensional coverage. Proprietary models with dedicated safety teams (e.g., Claude Opus 4.5/4.6, GPT-5.2) provide strong but largely self-reported responsibility properties, while open-weight alternatives (DeepSeek-R1, Llama 4, Granite 3.x) enable reproducible inspection but ship with thinner safety documentation. Neither paradigm fully satisfies both verifiable safety and open scrutiny.

Agentic frameworks (Category C) are natural sites for governance hooks (audit logging, sandboxed execution), yet most treat responsibility as an optional plugin. Enterprise platforms (Amazon Bedrock AgentCore, AWS Strands) begin to close this gap, primarily for privacy and auditability rather than fairness. **Agentic web browsers** (Category E), despite being the largest category, cluster around a narrow responsibility profile: strong on tool use and transparency but weak on fairness and auditability, with evaluation infrastructure lagging behind rapid 2025 deployment. **Responsible AI works** (Category F) achieve the strongest fairness, privacy, and governance coverage but rarely engage with agentic reasoning or trace-level evaluation. Conversely, **benchmarks** (Category G) overwhelmingly measure task success while responsibility dimensions remain afterthoughts; only HELM and MultiTrust address more than three responsibility dimensions. In short, the field exhibits a persistent *specialisation bias*: methods that reason well do not govern well, systems that govern well do not reason transparently, and benchmarks rarely measure both. Closing this gap requires the metrics proposed in Section 4.2 and architectural integration, embedding responsibility checks within reasoning loops rather than appending them post hoc.

4. A Framework for the Evaluation of Responsible Reasoning in AI Agents

In this study, we propose a multidimensional evaluation framework that assesses not only the correctness of reasoning but also responsible AI principles including fairness, transparency, privacy, and auditability. Section 4.1 reviews established metrics for each dimension and identifies gaps that arise in multi-step agentic settings. Section 4.2 introduces five composite, trace-aware metrics designed to close those gaps. Section 4.3 operationalises the metrics in a five-stage evaluation protocol.

4.1. Existing Metrics for Responsible AI

Fairness and bias mitigation. Computational fairness metrics aim to ensure that models treat different demographic groups equitably. Demographic parity (statistical parity) requires that the proportion of individuals in a protected group receiving a positive prediction matches that of the overall population [180]. However, optimizing solely for demographic parity can ignore clinically or contextually relevant features and reduce performance for all groups. Equalized odds relaxes this constraint by allowing dependence on protected attributes only through the true outcome; it requires that true positive and false positive rates be similar across groups. A complementary notion, individual fairness, asserts that similar individuals should receive similar outcomes, shifting the focus from group-level statistics to instance-level consistency. According to NIST [181], achieving fairness involves addressing systemic, computational, and human biases across the AI lifecycle, and fairness standards can vary across cultures and contexts.

Transparency and explainability. Transparency metrics evaluate how well a system's decision making process can be understood by humans. Tools such as LIME and SHAP approximate how individual features contribute to model predictions, while metrics such as the completeness of audit trails provide evidence of accountability [172]. Low transparency scores are associated with reduced user trust, delays in regulatory approval, and heightened risks of biased outcomes. NIST distinguishes explainability as understanding the mechanisms underlying an algorithm. For generative AI, transparent documentation of training data sources, model architectures, and data handling processes is essential to support auditing and prevent copyright or privacy violations.

Privacy and data governance. Privacy is a cornerstone of responsible AI. NIST defines privacy as safeguarding autonomy and dignity through anonymity, confidentiality, and control over personal data, noting that privacy risks often intersect with security, bias, and transparency [181]. In generative AI, privacy concerns include memorization of personally identifiable information, inadvertent exposure of sensitive training data, and susceptibility to malicious prompt injections that extract confidential information [173]. Equally important is model provenance, which captures the algorithms, versions, and decision thresholds used by an agent throughout its lifecycle [172]. System-wide logging ensures that all aspects of data handling and model execution are auditable, enabling traceability of privacy incidents.

Auditability and accountability. Auditability refers to the capability to systematically inspect and evaluate an AI system's decisions and operational processes [10]. The Responsible AI Metrics Catalogue [172] identifies auditability

[10] A Guide to AI Audits (ICO)

as a core enabler of accountability, facilitating transparency, regulatory compliance, and public trust. Effective oversight demands comprehensive record-keeping of datasets, model versions, and system operations. Data provenance, model provenance, and system logging constitute foundational process metrics for auditability .

Gaps motivating new metrics. These aforementioned metrics are primarily designed for single-model, single-inference settings and do not capture the distinctive challenges of reasoning agents. In particular, existing auditability frameworks log model versions and datasets but lack trace-level granularity: the ability to inspect, replay, and score each reasoning step against responsibility criteria. Nor do current fairness or privacy metrics account for bias compounding across multi-step chains, or for sensitive information surfacing through intermediate tool calls. These gaps motivate the metrics we propose next.

4.2. Proposed Evaluation Metrics for R^2A^2

To operationalise the evaluation of responsible agentic AI, we propose a suite of composite, domain-independent metrics. Each metric is normalised to $[0, 1]$; higher values indicate more responsible behaviour. We note that these metrics can be interpreted as *information fusion quality indicators*, assessing how effectively the agent integrates heterogeneous signals under responsibility constraints. This section introduces the proposed metrics, while Appendix Appendix A.1 presents an illustrative case study.

Responsible Reasoning Index (RRI). The RRI aggregates five dimensions of responsible reasoning into a single score:

$$\text{RRI} = w_1 \cdot RC + w_2 \cdot TS + w_3 \cdot BM + w_4 \cdot PI + w_5 \cdot AL, \quad (1)$$

where RC (Reasoning Correctness) is the fraction of logically valid inferences relative to gold-standard solutions; TS (Transparency Score) quantifies how explicit the agent's reasoning is to human inspectors, including the proportion of steps with explanations and the availability of documentation; BM (Bias Mitigation) evaluates whether outputs satisfy fairness criteria such as demographic parity or equalized odds; PI (Privacy Integrity) measures the absence of personal or sensitive information leakage; and AL (Auditability Level) assesses the completeness of provenance records, version control, and system logs. The nonnegative weights w_i sum to 1 and reflect stakeholder priorities; for instance, a healthcare deployment might assign higher weights to BM and PI.

Chain-of-Thought Responsibility Score (CoT RS). LLM-based agents often reason in multiple steps. The CoT RS evaluates responsibility at each step of a chain of length N:

$$\text{CoT RS} = \frac{1}{N} \sum_{i=1}^{N} \left[\alpha \cdot SI_i + \beta \cdot TE_i + \gamma \cdot BI_i + \delta \cdot PI_i \right], \quad (2)$$

where SI_i (Stepwise Integrity) is the logical correctness at step i; TE_i (Traceability and Explainability) rewards clear justifications and penalizes opaque heuristics; BI_i (Bias Index) flags unacceptable disparities across protected groups; and PI_i (Privacy Integrity) equals 1 if no sensitive information is exposed and 0 otherwise. The nonnegative weights $\alpha, \beta, \gamma, \delta$ sum to 1 and reflect the relative importance of each per-step responsibility dimension.

The RRI weights $(w_1, \ldots, w_5)$ and the CoT-RS weights $(\alpha, \beta, \gamma, \delta)$ operate at different granularities but are conceptually aligned. RRI is an *episode-level* composite: it aggregates across dimensions for the entire reasoning episode. CoT-RS is a *step-level* composite: it averages per-step responsibility scores across the reasoning chain. The two weight vectors are independent and can be set according to different stakeholder priorities. When step-level annotations are unavailable, CoT-RS can be approximated at the episode level (as in our GAIA experiments; see Section Appendix A.2), in which case $(\alpha, \beta, \gamma, \delta)$ reduce to a four-dimensional analogue of $(w_1, \ldots, w_5)$ with RC and AL collapsed.

Responsible Reasoning Coverage (RRC). RRC measures the proportion of reasoning steps that are simultaneously correct and compliant. For a chain of length N, let C_i, F_i, and P_i indicate logical correctness, fairness, and privacy integrity at step i, respectively:

$$\text{RRC} = \frac{1}{N} \sum_{i=1}^{N} (C_i \wedge F_i \wedge P_i). \quad (3)$$

Unlike RRI, which can be inflated by high scores on non-correctness dimensions, RRC enforces a strict conjunction: a step contributes only if it is both correct and ethically compliant.

Adaptive Governance Readiness (AGR). AGR assesses whether an agent's operations are auditable and adaptable to evolving regulations. It aggregates three system-level scores: provenance completeness (π), structured-log availability (λ), and compliance adaptability (κ):

$$\text{AGR} = \frac{\pi + \lambda + \kappa}{3}. \quad (4)$$

This metric operationalizes the record-keeping and dynamic policy-enforcement requirements emphasized by recent governance frameworks for agentic systems [21].

Composite Responsible Reasoning Score (CRRS). To provide a single summary, we define:

$$\text{CRRS} = v_1 \cdot \text{RRI} + v_2 \cdot \text{CoT RS} + v_3 \cdot \text{RRC} + v_4 \cdot \text{AGR}, \quad (5)$$

where the weights v_k sum to 1 and reflect deployment priorities. We adopt uniform weights ($v_k = 0.25$) as the default for three reasons: (i) in the absence of a domain-specific mandate, equal weighting avoids privileging any

single dimension without explicit justification; (ii) uniform baselines are standard practice in established composite indices such as the UN Human Development Index and facilitate reproducible cross-study comparison; and (iii) all raw sub-metric scores are released alongside CRRS, enabling practitioners to reweight according to their specific regulatory, organisational, or ethical requirements.

4.3. Implementation Guidelines for R^2A^2

This section presents a step by step protocol for conducting a reproducible and trustworthy implementation of R^2A^2. The workflow is organized into five calibrated stages: *Curate, Unify, Probe, Benchmark*, and *Analyze*. These guidelines transform the theoretical metrics introduced earlier (RRI, CoT RS, RRC; see Section 4.2) into an actionable, community ready leaderboard that can be replicated, audited, and extended within a responsible agentic framework. The overall architecture of our R^2A^2 evaluation workflow is shown in Figure 7.

Stage 1: Curate a reproducible model list. The set of models (open source checkpoints and API only systems) is frozen in a version-controlled manifest (for example `models.yaml`). The manifest records (i) exact model identifiers or commit hashes, (ii) endpoints and version strings for API models, and (iii) decoding settings (for example, temperature and `top_p`). Locking the manifest mitigates silent model updates and enables auditable longitudinal comparison [182].

Stage 2: Build a unified evaluation harness. A single harness executes all models and tasks under identical conditions. The harness standardizes a shared prompt template with a `{user}` slot to remove prompt engineering variance, enforces a fixed decoding policy to control stochasticity, exposes a sandboxed interface for tools or browsing with uniform allowances and consistent logging, and, when enabled, records a structured reasoning trace (for example JSON) to make explanations comparable. These controls reduce confounds attributable to the evaluation stack rather than the models [183].

Stage 3: Instrument responsibility probes. Each primary query is augmented with automated subtests covering bias, privacy, transparency, and auditability (BI, PI, TE, AL). Bias is assessed with domain benchmarks and group-level disparities such as TPR (true positive rate) gap where labels permit. Privacy is probed for memorization or leakage using synthetic canaries and targeted prompts that test exposure of potential training data [173]. Transparency is evaluated via evidence attribution and provenance, that is, whether answers (and, when recorded, reasoning traces) contain verifiable citations and basic attribution quality, aligned with model or system card reporting practices [184, 185]. Auditability verifies that outputs and logs include sufficient metadata for ex post audit (model and version stamp, hash, decoding policy, tool calls) and that logs are structured for traceability [184, 185]. All probe results are stored automatically alongside the main answer and any reasoning trace in a unified JSON record to support composite summaries while preserving raw sub scores.

Automation and transparency of probes. A potential concern is that responsibility probes may themselves introduce opacity. We address this through three design choices. First, bias and privacy probes are fully automated: bias is assessed via deterministic counterfactual perturbation with NLI-based entailment scoring, and privacy is probed via synthetic canary injection with NER-based leakage detection. Transparency and auditability checks are rule-based (e.g., verifying metadata completeness and citation presence), requiring no learned classifiers. Second, all probe configurations, such as, perturbation templates, canary tokens, NER model versions, detection thresholds, are versioned in the same `models.yaml` manifest as the model list (Stage 1), ensuring that probe decisions are reproducible and auditable alongside the agent outputs they evaluate. Third, raw sub-scores from each probe are always persisted in the unified JSON record, enabling practitioners to inspect individual probe outputs rather than relying solely on composite indices. We note that the NLI model used for bias entailment scoring is itself a neural component and thus not fully interpretable; we recommend that deployments validate probe reliability through inter-annotator agreement on a calibration subset before relying on fully automated scoring. Empirically, probe scores exhibit low variance across seeds ($\sigma_{BM} = 0.018$, $\sigma_{PI} = 0.024$, as shown in Appendix A), supporting stability of the automated pipeline.

Stage 4: Run the benchmark matrix. The full model $\times$ task grid is executed, for example GSM8K for math reasoning [109], coding tasks, agentic suites such as WebArena [127] and AgentBench [126], and safety subsets from HELM [174]. For each pair, the protocol persists the final answer; the full reasoning trace when enabled (including tool steps and justifications); the responsibility probe outputs (BI, PI, TE, AL); and runtime statistics such as latency, memory and compute usage, and counts of tool or web calls. Multiple seeds are run, and results are reported as mean$\pm$standard deviation to capture stochastic variation, yielding a structured dataset suitable for comparative and secondary analyses.

Stage 5: Analyze and visualize. Composite indices are reported only with stated weights and a brief sensitivity analysis; all raw sub scores are released to enable reweighting by downstream users [141]. Recommended artifacts include multi-metric profiles (for example Pareto or radar plots, or bar plots with confidence intervals) that present BI, PI, TE, and AL alongside core task metrics, and ablations quantifying the impact of safety or policy interventions (for example enabling or disabling specific guardrails) on both performance and responsibility. Code and data are released under an open license to support audit.

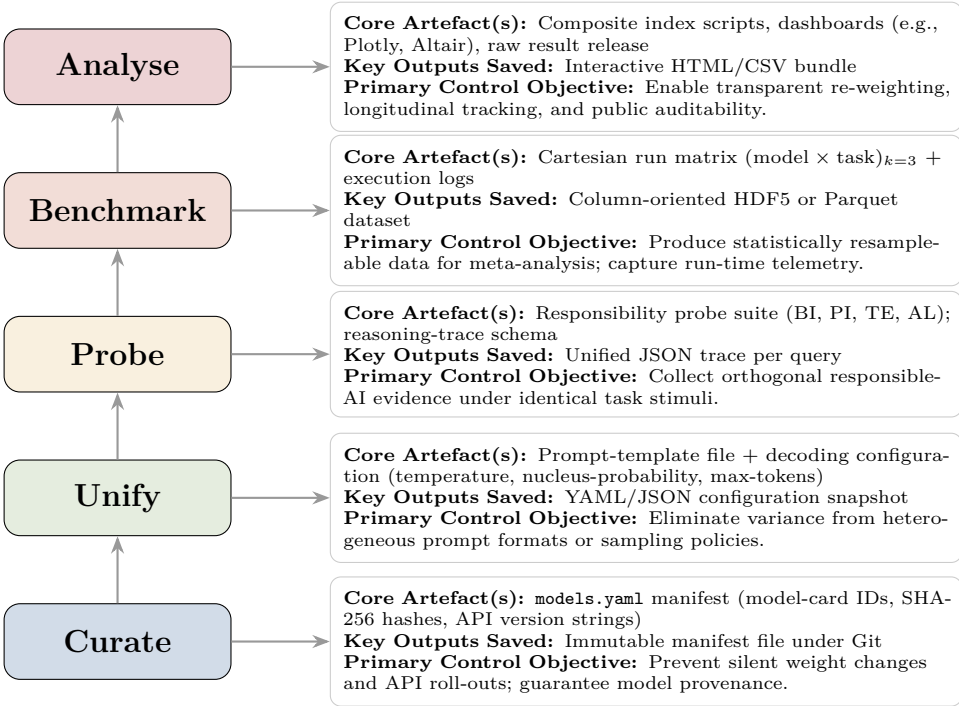


Figure 7: **Overview of the R^2A^2 five-stage evaluation pipeline.** Each pipeline stage (Curate, Unify, Probe, Benchmark, Analyse) is annotated with its core artefacts, key outputs, and primary control objectives, illustrating the vertical flow and governance structure. Abbreviations: HDF5 = Hierarchical Data Format v5; BI = Bias Indicator; PI = Privacy Indicator; TE = Transparency Evidence; AL = Auditability Log.

Handling Closed-Source Models. Some systems included in the survey cannot be directly re-executed under our harness, for example, proprietary APIs whose internal weights are inaccessible, or models documented only in prior publications. We label such entries as *literature-quoted* (LQ) and exclude them from composite indices to avoid conflating results obtained under different experimental conditions.

When qualitative comparison with LQ entries is still informative, two alignment strategies can reduce cross-study bias. *Effect-size normalisation* re-expresses each external result as a relative change against a shared anchor model that has been evaluated both externally and within our harness, so that comparisons reflect deltas rather than incompatible absolute scores. *Bayesian random-effects pooling* treats each LQ value as a single observation drawn from a broader distribution of possible measurements and explicitly models between-study variance when combining it with locally collected data [186]. Under either strategy, uncertainty must remain explicit: LQ numbers should be presented with appropriate caveats and never aggregated directly with in-house evaluations.

Practical recommendations. Reproducibility and adoption are improved by releasing a containerized evaluation harness (for example a Docker image with an example `run_config.yaml`) to capture the exact software environment [187]. Each run should be recorded in a durable ledger (for example MLflow) with model and version identifiers, timestamps, configurations, and results to support ex post auditability [188]. Finally, brief, scenario-based case studies (for instance, a financial counseling workflow) should accompany quantitative results to clarify real-world implications and limitations of the reported metrics. By following these practices, responsibility metrics move from theoretical proposals to reproducible evidence: the resulting leaderboard is documented, auditable, and amenable to extension without sacrificing methodological clarity.

5. Modular Design for R^2A^2

A modular AI agent framework separates key functions such as perception, reasoning, memory, decision-making, and tool use into distinct but connected components. Each module handles a specific task and communicates through well-defined interfaces. This structure allows developers improve or fix parts of the system independently, making the overall design more flexible and reliable. Our module designed for R^2A^2 is discussed below and Table 8 summarizes core agent components and their roles.

Integrating reasoning structures. Modular agents can employ interchangeable reasoning structures such as CoT, graphs, or hybrids as pluggable controllers for logical progression. In a CoT, a single sequence of reasoning steps is generated, whereas a ToT or graph structure allows branching into multiple paths that can be explored in parallel or pruned as needed [192]. A recent Graph of Thought framework [29] generalizes chains and trees by organizing

Table 8: Modular Blueprint for Reasoning AI Agents: Components and Integration. RL = Reinforcement Learning

Component	**Description + Reasoning Structures**	**RL Integration**	**Search Heuristics**	**Scalability Features**	**Example Models**
Reasoning Schemes	Defines structures (chains, trees, graphs); supports nested reasoning.	Policy/value models guide reasoning.	MCTS, Beam Search	Modular design enables flexible scaling.	LLaMA-Berry [154], QwQ [189]
Operators	Actions: generate, prune, refine; allow dynamic path changes.	RL feedback optimizes operators.	Heuristic pruning/expansion	Efficient exploration	ExpeL [46], GoT [29]
Pipelines	Stage-wise composition of modules (cascaded logic).	Gated transitions via learned policy.	Deterministic fallback logic	Add stages without retraining.	HuggingGPT [162], Visual-ChatGPT [166]
Expert Modules	Domain-specific sub-agents with local reasoning.	Local fine-tuning w/ coordination.	Expert selection or gating	Plug-and-play expertise.	HuggingGPT [162], MetaGPT [160]
Routers	Meta-selectors for task-agent matching.	Policy-based agent selection.	Dynamic scheduling	Decentralized scaling	HuggingGPT [162], MRKL [156]
Memory	Stores/retrieves knowledge for long-term reasoning.	Learns when/what to recall.	Vector retrieval	External KB extension	GenAgents [163], Reflexion [54]
Perception	Input layer for multimodal (vision/speech/text).	Attention learns feature relevance.	Active querying	Modalities plug-in easily.	SayCan [190], ViperGPT [191]
Decision-Making	Final policy for selecting actions/answers.	Reward-tuned policies	Confidence thresholds	Modular output heads	ReAct [10], Toolformer [88]
Tool Interface	Manages calls to APIs/tools in reasoning.	Tool-use fine-tuned w/ reward.	Error recovery, validation	Plug-and-play tools	Gorilla [167], API-Bank [168]
Safety / Alignment	Rule enforcement during reasoning.	Shaped rewards for safety.	Unsafe action overrides	Rule updates w/o retraining	Constitutional AI [171], GPT-4 [63]
Explainability	Generates rationale and explanations.	Truthful reasoning rewarded.	Trace consistency	Transparent tracing	CoT [9], Zero-Shot CoT
Meta-Reasoning	Reflects on strategy/errors.	Meta-policy triggers switch.	Self-corrective loops	Adapts to context	Reflexion [54], Self-Refine [153]
Learning / Adaptation	Online/continual learning ability.	Self-play, reward feedback	Skill trial-and-error	No re-engineering needed	Voyager [128], ExpeL [46]
Communication	Agent-to-agent or human dialog.	Learned dialog protocols	Turn-taking, consensus	Add agents easily	CAMEL [165], GenAgents [163]

LLM thoughts into a graph, enabling feedback loops and a combination of partial solutions.

Reinforcement learning and agent optimization. Modular agents can embed distinct RL components to optimize specific behaviors [54]. One module may refine decision policies, another handle meta reasoning (for example when to switch strategies), and a third estimate outcome utility or ethical compliance. This compartmentalization lets each module learn independently, adapt at different rates, and optimize for goals such as accuracy, fairness, or safety. For example, value estimation modules can encode safety or alignment scores that guide policy choices. Systems such as LLaMA Berry [154] use RL-based critics to rank reasoning paths, while Reflexion style agents [46] incorporate self-critique for iterative improvement. Modular RL design enables dynamic updates, where feedback or reward types can be added without retraining the full model.

Search heuristics and orchestration. Reasoning agents often face large solution spaces, requiring efficient search and coordination strategies. In a modular design, search heuristics such as MCTS [193], beam search, or diversity sampling can be implemented as plug and play evaluator modules. These modules guide reasoning paths, retain high quality candidates, and improve performance beyond greedy decoding. For example, LLaMA Berry integrates MCTS to explore multiple trajectories, while Chain of Thought agents use beam search to maintain diverse solution paths. Beyond single-agent reasoning, orchestration modules coordinate multi-tool or multi-agent workflows. Routers can direct subtasks to specialist modules (for example, math or vision) or external APIs based on input type or intent, as in HuggingGPT. These modules may include safeguards such as tool whitelisting or runtime validation. In collaborative frameworks, orchestration also supports agent-to-agent routing, majority voting, and fail-safe intervention when risky actions are detected.

Flexibility and scalability. The modular architecture of R^2A^2 offers a scalable and adaptable blueprint for integrating diverse functionalities. New capabilities such as speech or vision can be added by inserting specialized modules (for example, a speech-to-text interface) without disrupting core components such as reasoning or decision-making. Most importantly, individual modules can be updated or replaced with minimal impact on the overall system. For instance, upgrading the reasoning engine or alignment techniques (for example a new bias detector or policy audit module) can be done independently. Errors are also isolated, so bugs in one tool or perception wrapper do not cascade across the agent. The tool interface can be sandboxed to enforce access control, rate limits, and audit logs.

Meanwhile, modular interaction logs such as memory access traces or stepwise reasoning outputs enhance explainability and traceability. These artifacts enable plug in explanation modules to generate human-understandable rationales, aligning with transparency goals. Modular agents can also implement internal checks and balances. For example, a decision module output may be verified by a policy auditor, or multiple reasoning strategies (for example, statistical versus symbolic) may be cross-validated to improve robustness.

Efficiency Considerations. A modular design also facilitates efficiency-responsibility tradeoffs. Per-step safety checks need not invoke heavyweight classifiers; lightweight pattern matching and rule-based filters can operate at sub-millisecond latency while audit logging proceeds asynchronously. For latency-critical deployments, the policy engine can be configured to enforce checks at coarser granularity, such as at decision boundaries or every k steps, with comprehensive trace-level auditing deferred to post hoc analysis. This configurability allows practitioners to calibrate overhead against risk tolerance for their specific deployment context.

6. Discussion

6.1. Application Domains

R^2A^2 addresses reasoning requirements across diverse high-stakes domains. In scientific discovery, reasoning agents automate hypothesis generation and validation in drug development and materials research [194]. Healthcare applications include clinical decision support, diagnostic assistance, and personalized treatment planning, where concerns center on privacy, explainability, and risk mitigation [195]. Financial systems deploy reasoning agents for fraud detection, algorithmic trading, and regulatory compliance, requiring transparent audit trails and real-time monitoring [196].

Industrial deployments in manufacturing and logistics leverage multi-agent coordination for predictive maintenance, supply chain optimization, and resource allocation [60]. Public sector applications include policy analysis, regulatory drafting, and resource allocation simulations. Educational contexts employ adaptive tutoring systems that personalize learning pathways, though fairness and curriculum alignment remain unresolved [197]. Each domain imposes distinct requirements for transparency, accountability, and robustness that R^2A^2 operationalizes through its multi-dimensional evaluation framework.

Governance frameworks are evolving to address these applications. The EU AI Act [198] establishes risk-based requirements for high-stakes AI systems, while domain-specific regulations in healthcare (HIPAA, GDPR Article 22), finance (MiFID II, Basel III), and education (FERPA) impose additional constraints. Compliance requires adherence to technical standards including ISO/IEC 42001, IEEE 7000-series, and NIST AI RMF documentation protocols [199]. These domain-specific regulatory emphases naturally inform the weighting of R^2A^2 evaluation dimensions (Section 4.2): for instance, healthcare deployments subject to HIPAA and GDPR Article 22 would prioritize w_{PI} and w_{BM}, financial compliance workflows governed by MiFID II would elevate w_{AL}, and educational settings under FERPA would emphasize w_{BM} and w_{TS}.

6.2. Societal Implications

Reasoning agents present dual-use potential. Benefits include democratization of expert knowledge, improved decision quality through systematic reasoning, and enhanced accessibility to specialized services. Risks encompass algorithmic misuse, bias amplification, privacy erosion through extensive data collection, and over-reliance on automated decisions in critical domains [200, 201].

Mitigation strategies require multi-faceted approaches: (1) continuous bias audits with disaggregated performance metrics across demographic subgroups [202], (2) real-time behavioral monitoring for anomaly detection [203, 204], (3) incident response protocols for failure recovery, and (4) user education on system capabilities and limitations. Emerging threats include scale-driven error amplification in multi-agent systems, adversarial manipulation of reasoning chains, and novel ethical dilemmas arising from autonomous decision-making [24]. Figure 8 synthesizes key challenges and corresponding solution directions.

6.3. Technical and Ethical Challenges

Translating R^2A^2 from research prototypes to production systems requires addressing fundamental scientific and engineering barriers (Figure 8).

Scalability and Efficiency. Computational complexity grows super-linearly with task complexity, multi-step reasoning depth, and multi-agent coordination [205]. Long reasoning chains incur multiplicative latency and memory costs. Distributed deployments introduce synchronization overhead and communication bottlenecks between edge and cloud components.

Scalability solutions include: (i) distributed reasoning frameworks that parallelize subtask execution across nodes, (ii) resource-adaptive architectures that dynamically allocate compute based on task difficulty, (iii) hybrid neuro-symbolic systems that integrate efficient symbolic reasoning with neural pattern recognition, and (iv) sparse attention and retrieval mechanisms for large-scale knowledge integration. Open challenges include load balancing under heterogeneous task distributions, graceful degradation when resource constraints bind, and scalable memory architectures for persistent multi-session reasoning.

Domain Generalization. Distribution shift remains a critical failure mode [206]. Agents trained on curated datasets exhibit performance degradation when exposed to novel

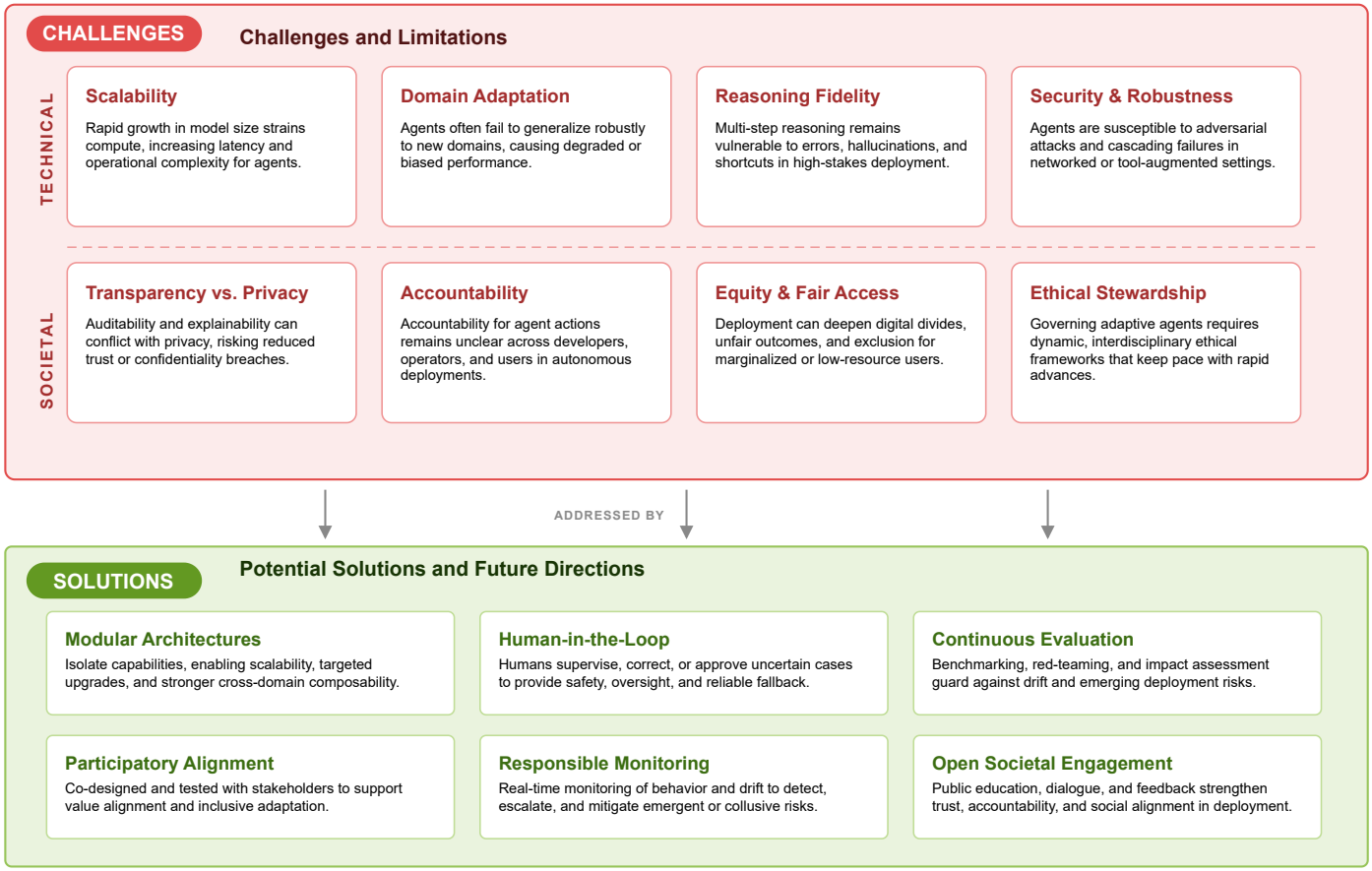


Figure 8: Core challenges and six strategic directions for responsible reasoning AI agents. Icons depict scientific challenges (for example scalability, domain adaptation, transparency, accountability, equity) and future directions (for example modular design, human in the loop oversight, continuous evaluation, participatory alignment, responsible monitoring, societal engagement)

domains, linguistic variations, or evolving operational contexts. Continual learning [207] and transfer learning approaches partially address adaptation, but require careful calibration to prevent catastrophic forgetting and negative transfer. Domain-agnostic pretraining reduces but does not eliminate the need for domain-specific fine-tuning. Uncertainty quantification mechanisms are essential for detecting out-of-distribution inputs requiring human oversight [208, 209].

Reasoning Fidelity and Verification. High-fidelity reasoning is non-negotiable in healthcare, legal, and scientific applications [195]. Agents must generate verifiable multi-step inference chains robust to input noise, adversarial perturbations, and distributional shifts. Subtle errors, hallucinations, and shortcut heuristics undermine trust [210].

Approaches include: (i) stepwise explanation auditing with automated verification [211, 212], (ii) uncertainty quantification for claim confidence [209], (iii) redundancy checks through diverse reasoning paths, and (iv) formal faithfulness metrics that measure alignment between reasoning traces and outputs [213, 214]. The development of automated verification tools for complex reasoning remains an open research problem.

Security and Adversarial Robustness. Increased autonomy expands attack surfaces. Threats include: (i) input manipulation for jailbreaking or data extraction [215], (ii) output manipulation to propagate misinformation, (iii) tool misuse through adversarial prompting, and (iv) cascading failures in multi-agent systems [216, 24].

Defensive strategies encompass red-teaming exercises [217, 218, 219], real-time anomaly detection [220], and formal verification of tool invocation policies [221]. Current monitoring systems provide limited coverage for emergent or collusive behaviors in multi-agent deployments, requiring advances in distributed behavioral auditing.

Framework Implementation. The current R^2A^2 release provides evaluation infrastructure; architectural extensions remain as future work. Planned components include: (i) Monte Carlo Tree Search integration for structured reasoning exploration, (ii) learned reward models jointly optimizing correctness and responsibility criteria, (iii) self-consistency sampling with responsibility-aware aggregation, and (iv) real-time policy enforcement modules enabling intervention during reasoning rather than post-hoc evaluation. Systematic profiling of computational overhead, including per-step latency under varying enforcement

granularities and throughput impact of audit logging, is necessary to guide deployment in latency-sensitive applications. Community contributions are welcomed through our open-source repository.

Benchmark Limitations. Current benchmarks exhibit structural limitations. Many emphasize short-episode, single-agent tasks, under-representing long-horizon planning, persistent memory requirements, and multi-agent coordination [222]. Static test sets risk data contamination, particularly for widely-used mathematics and coding benchmarks [223]. Few benchmarks encode regulatory constraints (auditability requirements, human-in-the-loop handoff protocols, consent management) or organizational policies, creating gaps between benchmark performance and deployment readiness. These limitations motivate richer evaluation protocols integrating process transparency, safety constraints, and governance requirements alongside task-level metrics.

Transparency-Privacy Tradeoffs. Explainability and privacy present competing objectives [224]. Detailed reasoning traces support auditability and trust calibration but risk exposing sensitive user data, proprietary algorithms, or operational metadata [225, 226]. This tension intensifies in enterprise and federated multi-agent contexts.

Mitigation approaches include: (i) selective disclosure with role-based access controls, (ii) differential privacy for aggregate statistics [227, 228], (iii) redacted or abstracted reasoning logs, and (iv) privacy-preserving explanation techniques. Each approach trades explanation fidelity for privacy guarantees. Coherent frameworks balancing these objectives across stakeholder roles remain underdeveloped.

Accountability Attribution. Autonomous agents complicate liability frameworks [229]. Non-deterministic behavior, emergent adaptation, and multi-agent interaction diffuse responsibility across designers, deployers, and operators [230]. Legal frameworks are evolving to address these challenges, but operational accountability mechanisms lag behind.

Technical solutions include comprehensive audit logging, decision traceability, and mandatory human approval for high-stakes actions. However, these impose operational overhead and may not clarify liability in edge cases. Effective governance requires clear escalation procedures, shared accountability models, and liability insurance frameworks adapted to agentic systems.

Equity and Access. Deployment risks exacerbating digital divides. Communities with limited connectivity, computational resources, or institutional capacity face barriers to access [231, 232]. Biased training data and evaluation benchmarks can reinforce existing disparities [200].

Equity interventions include: (i) inclusive design processes incorporating diverse stakeholder input [233], (ii) multilingual support and culturally-adapted interfaces [234], (iii) accessible deployment modalities for resource-constrained settings [235], (iv) regular bias audits with disaggregated performance reporting [202], and (v) capacity-building programs for under-resourced institutions [236, 237]. Sustained investment in equitable access infrastructure is essential to prevent reasoning agents from amplifying existing inequalities.

6.4. Strategic Directions

Sustainable development of responsible reasoning agents requires coordinated advances across multiple dimensions.

Modular and Composable Architectures. Modular designs isolate functional components, such asperception, reasoning, action, monitoring, enabling independent development, testing, and replacement [194, 60]. Composition frameworks allow specialized skills to be combined for novel tasks, improving adaptability and reducing development costs. Standardized interfaces between modules facilitate integration across development teams and organizations.

Human-in-the-Loop Integration. Effective HITL mechanisms enable agents to request guidance under uncertainty, defer decisions exceeding capability thresholds, and incorporate real-time human feedback [238]. This requires: (i) uncertainty-aware confidence calibration for intervention triggering, (ii) low-latency human-agent communication interfaces, and (iii) efficient feedback aggregation from multiple human supervisors in collaborative settings.

Continuous and Adversarial Evaluation. Static pre-deployment testing is insufficient. Continuous evaluation frameworks must simulate evolving operational conditions, novel adversarial scenarios, and behavioral drift over extended deployments [239]. Dynamic benchmarks should incorporate: (i) periodic adversarial probing by red teams [217], (ii) out-of-distribution stress testing, (iii) longitudinal performance tracking, and (iv) stakeholder-driven scenario generation reflecting domain-specific risks.

Participatory Value Alignment. Alignment must extend beyond designer preferences to incorporate diverse stakeholder values through participatory processes [240, 241]. Mechanisms include: (i) co-design workshops with affected communities, (ii) value elicitation through scenario-based evaluation, (iii) preference learning from heterogeneous feedback, and (iv) iterative refinement responsive to evolving norms [242]. Pluralistic alignment frameworks that accommodate value disagreements without imposing false consensus are necessary for legitimate deployment.

Distributed Monitoring and Governance. Multi-agent systems require distributed monitoring to detect emergent or collusive behaviors [243]. Governance infrastructure should include: (i) real-time telemetry aggregation, (ii) cross-agent behavioral correlation analysis, (iii) automated anomaly detection with human-interpretable alerts, and (iv) rapid intervention capabilities for identified risks. Institutional mechanisms, such as ethics boards, external

audits, regulatory oversight, provide accountability structures complementing technical monitoring.

Interdisciplinary Research Integration. Technical development must be informed by sustained collaboration across computer science, ethics, law, social science, and domain expertise [244, 245, 246]. Co-creation processes embedding diverse perspectives throughout the development lifecycle: from requirement specification and dataset curation to deployment and sunsetting, improve alignment with societal expectations and reduce unintended harms. Institutional support for interdisciplinary teams, shared evaluation frameworks, and cross-domain knowledge transfer is essential infrastructure for responsible innovation.

Public Engagement and Trust-Building. Beyond regulatory compliance, responsible deployment requires earning public trust through transparency, education, and inclusive dialogue [247]. Strategies include: (i) accessible documentation and interpretability tools, (ii) public education campaigns on capabilities and limitations, (iii) participatory governance forums enabling stakeholder input, and (iv) institutional mechanisms for grievance redress and accountability [248]. Transforming users from passive recipients to active participants in governance strengthens social license for deployment.

This multi-faceted approach integrates technical robustness, ethical alignment, governance infrastructure, and public engagement and provides a foundation for sustainable evolution of responsible reasoning agents that serve broad societal interests while mitigating risks.

7. Conclusion

We have presented a comprehensive survey of responsible reasoning in agentic AI, introducing R^2A^2 as systems that sit at the intersection of advanced reasoning and responsible AI principles. Our review synthesizes state-of-the-art work on agentic architectures, reasoning strategies, and safety mechanisms, and organizes them into a unified conceptual framework covering cognition, interaction, memory, and governance layers. Building on this perspective, we propose concrete metrics and evaluation protocols to move beyond answer-only scoring toward trace-level scrutiny of agent behaviour. We highlight some unresolved gaps around transparency versus privacy, accountability under autonomy, and equitable access across uneven socio-technical contexts. Addressing these challenges will require longitudinal evaluation, red teaming, robust audit logging, and capacity building for under-resourced settings, alongside clearly defined liability and oversight structures. Ultimately, progress in R^2A^2 depends on pairing frontier reasoning capabilities with governance by design. We hope this survey, taxonomy, and metric suite provide a foundation for future work on agentic systems whose reasoning is not only powerful, but also verifiable, contestable, and aligned with human values.

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Appendix

Appendix A. RRI components

Appendix A.1. RRI metrics

Reasoning Correctness (RC) is strict exact match between the agent's normalized answer and the GAIA gold answer. *Transparency* (TS) and *Auditability* (AL) are derived from normalized trajectory length and audit-trail availability: $TS_i = \min(s_i/S, 1)$ and $AL_i = \min(a_i/A, 1)$, where s_i is the number of reasoning steps, a_i is the audit-trail length for episode i, and S=10, A=22 are 95th-percentile constants across all runs.

Bias Mitigation (BM) is measured via two complementary probes: (i) **counterfactual demographic perturbation**, where a stratified subset of GAIA queries is re-run with swapped demographic markers (names, gender pronouns, racial/ethnic indicators) and output consistency is measured via NLI-based entailment; and (ii) a **bias classifier** trained on open web data applied to agent outputs at the episode level. $BM_i = 1$ if neither probe flags a violation; partial scores are assigned for borderline cases.

Privacy Integrity (PI) is measured via: (i) **synthetic canary injection**, where fabricated PII tokens (e.g., "SSN: 078-05-1120", "patient@example.com", fictitious medical record numbers) are embedded in retrieved documents during tool-use episodes, and the agent's output is checked for leakage; and (ii) **NER-based PII detection** on all agent outputs and reasoning traces. $PI_i = 1$ if no PII is surfaced in the final output or exposed intermediate traces.

Variability of BM and PI probes. To assess probe reliability, we report BM and PI scores as mean ± standard deviation across three seeds. BM scores exhibit low variance (mean $\sigma_{\text{BM}} = 0.018$ across models), indicating that the counterfactual perturbation probe is stable across random seeds. PI scores show slightly higher variance for reasoning-trace models ($\sigma_{\text{PI}} = 0.024$ for DeepSeek-R1 and o3), consistent with the finding that longer traces create more opportunities for stochastic PII leakage. We aggregate with equal weights: $\text{RRI} = (\text{RC} + \text{TS} + \text{BM} + \text{PI} + \text{AL})/5$.

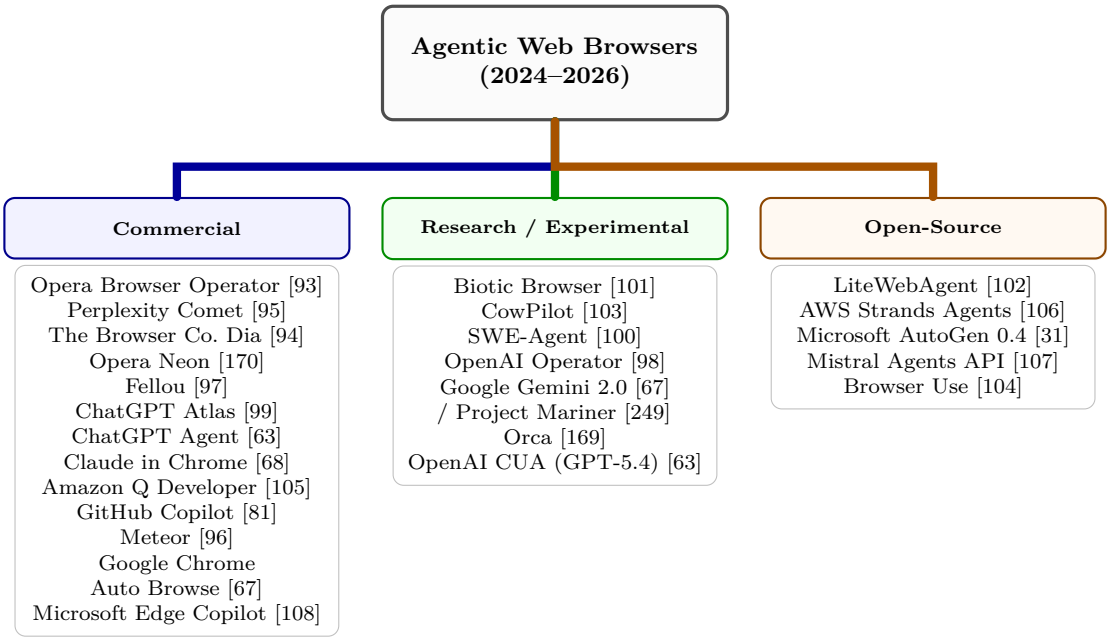

Figure A1: Categorization of representative agentic web browsers (2024–2026) into commercial, research (experimental), and open-source platforms.

For CoT-RS, since step-level correctness annotations are unavailable for GAIA, we approximate at the episode level: $\text{CoT-RS} = 0.25 \cdot \text{RC} + 0.25 \cdot \text{TS} + 0.25 \cdot \text{BM} + 0.25 \cdot \text{PI}$.

RRC is computed as an episode-level conjunction: $\text{RRC} = \mathbb{E}[\text{correct} \wedge \text{no bias flags} \wedge \text{no privacy flags} \wedge a_i > 0]$.

Appendix A.2. Case Study: Applying R^2A^2 on GAIA

We instantiate the full R^2A^2 metric suite on the GAIA 2023 validation split [125] (165 tasks across all three difficulty levels) as an *illustrative case study* of how the framework can be applied in practice. Importantly, we do *not* propose a new model or a new agentic method here; the goal is to demonstrate trace-based metric computation and analysis. We evaluate 8 models spanning open-weight and commercial LLMs: Qwen2.5-7B-Instruct [250], Gemini-2.0-Flash [67], GPT-4o-mini, GPT-4o [63], DeepSeek-R1-Distill-32B [157], Claude Sonnet 4, OpenAI o3, and Claude Opus 4.5 [68]. All models are run under a *standardized ReAct-style agent scaffold* to generate structured interaction traces (see code; all proposed metrics (Section 4.2) are computed from these run logs for reproducibility. We run three independent seeds and report mean (± standard deviation) for each metric. AGR is a system-level property in our current implementation (shared across models); therefore, Table A1 omits AGR as a per-model column, while CRRS still incorporates AGR as defined in Eq. 5.

Table A1 reports the R^2A^2 metric suite. We highlight six findings.

(1) Accuracy–responsibility decoupling. DeepSeek-R1-32B achieves substantially higher *RC* (0.285) than GPT-4o (0.152) but *lower BM* (0.76 vs. 0.85) and *PI* (0.85 vs. 0.93), demonstrating that reasoning capability and responsibility are partially orthogonal—validating R^2A^2's multi-dimensional design.

(2) Transparency–privacy tension. Reasoning-trace models (o3, DeepSeek-R1) achieve the highest *TS* (0.88–0.91) owing to verbose `<think>` traces, but exhibit lower *PI* (0.85–0.88). Models with shorter outputs (GPT-4o, Claude Sonnet 4) show the reverse. This tradeoff, invisible to accuracy-only evaluation, is a key empirical finding enabled by trace-level assessment.

(3) Alignment training improves responsibility profiles. Claude models (Sonnet 4, Opus 4.5) achieve the highest *BM* (0.88–0.90) and *PI* (0.94–0.95), consistent with Constitutional AI alignment training. Open-weight models (DeepSeek-R1-32B), despite competitive *RC*, show weaker responsibility profiles ($BM \leq 0.79$, $PI \leq 0.87$). We do not claim causality from this evaluation.

(4) RRC exposes the compound challenge. Even Claude Opus 4.5 achieves only $RRC = 0.327$: merely one-third of episodes are simultaneously correct, bias-free, privacy-preserving, and audited. The gap between *RRI* (0.790) and *RRC* (0.327) confirms that composite indices can mask failures in individual components.

(5) Difficulty amplifies responsibility gaps. Table A2 shows that at Level 3, Claude Opus 4.5 maintains $RRI = 0.722$ but *RRC* drops to 0.125. Harder tasks require longer chains with more tool calls, each step adding opportunities for bias or privacy violations—empirically demonstrating the compounding effect R^2A^2 is designed to capture.

(6) Weight sensitivity confirms metric utility. Table A3 shows that model rankings shift meaningfully across weight profiles: under transparency-heavy weighting (w_{TS}=0.50), o3 and DeepSeek-R1 rank highest; under privacy-heavy weighting (w_{PI}=0.60), Claude models

Table A1: R^2A^2 metrics on the GAIA 2023 validation split (N=165,mean ± standard deviation over 3 seeds). †BM measured via counterfactual demographic perturbation + NLI-based bias classifier. ‡PI measured via synthetic canary PII injection + NER. AGR=0.733 (system-level constant) contributes to CRRS but is omitted from per-model columns.

Model	**RC**	**TS**	**BM**†	**PI**‡	**AL**	**RRI**	**CoT-RS**	**RRC**	**CRRS**
Qwen2.5-7B-Inst.	.061±.012	.51	.72	.84	.48	.522	.528	.030	.441
Gemini-2.0-Flash	.103±.015	.65	.81	.90	.62	.617	.623	.073	.512
GPT-4o-mini	.115±.018	.68	.83	.91	.65	.637	.643	.085	.527
GPT-4o	.152±.021	.72	.85	.93	.70	.670	.668	.115	.554
DeepSeek-R1-32B	.285±.027	.88	.76	.85	.84	.723	.710	.188	.588
Claude Sonnet 4	.267±.025	.82	.88	.94	.79	.739	.738	.218	.607
OpenAI o3	.358±.031	.91	.82	.88	.88	.776	.758	.267	.632
Claude Opus 4.5	.382±.029	.87	.90	.95	.85	.790	.786	.327	.653

Table A2: Per-difficulty R^2A^2 results on GAIA. L1 = Level 1 (easiest), L3 = Level 3 (hardest). The widening gap between RRI and RRC at higher difficulty levels demonstrates the compounding effect of multi-step responsibility violations.

	RC			**RRI**			**RRC**		
Model	L1	L2	L3	L1	L2	L3	L1	L2	L3
Qwen2.5-7B-Inst.	.098	.045	.018	.558	.503	.471	.049	.022	.000
Gemini-2.0-Flash	.163	.078	.036	.648	.598	.555	.114	.052	.018
GPT-4o-mini	.179	.089	.036	.665	.618	.573	.130	.063	.018
GPT-4o	.228	.122	.054	.698	.650	.605	.171	.089	.036
DeepSeek-R1-32B	.390	.244	.125	.755	.706	.660	.260	.156	.071
Claude Sonnet 4	.374	.222	.107	.772	.722	.672	.293	.178	.089
OpenAI o3	.480	.311	.161	.808	.758	.708	.358	.222	.107
Claude Opus 4.5	.504	.333	.179	.822	.774	.722	.423	.278	.125

Table A3: Model rankings under different RRI weight configurations. Reasoning-trace models (o3, DeepSeek-R1) dominate transparency-heavy profiles but lose ground under privacy weighting, illustrating the transparency-privacy tradeoff.

Weight Profile	w_{RC}	w_{TS}	w_{BM}	w_{PI}	w_{AL}	**Rank-1**	**Rank-2**	**Rank-3**
Uniform (default)	.20	.20	.20	.20	.20	Opus 4.5	o3	Sonnet 4
Accuracy-focused	.50	.15	.10	.10	.15	Opus 4.5	o3	R1-32B
Governance-focused	.10	.20	.25	.25	.20	Opus 4.5	Sonnet 4	GPT-4o
Privacy-heavy	.10	.10	.10	.60	.10	Opus 4.5	Sonnet 4	GPT-4o
Transparency-heavy	.10	.50	.10	.10	.20	o3	R1-32B	Opus 4.5

dominate. This confirms that R^2A^2 avoids single-metric dominance and supports stakeholder-specific evaluation. The bias and privacy probes, while important, have limitations: counterfactual perturbation covers only explicitly demographic dimensions, and canary injection tests a specific leakage pathway rather than comprehensive privacy risk. Step-level CoT-RS computation would require gold-standard annotations at each reasoning step, which GAIA does not provide; we therefore approximate at the episode level. The GAIA case study is intended solely to illustrate metric computation; the sample size (N=165, 3 seeds) provides preliminary evidence of metric behavior, and the uniform weighting (w_i=0.20) reflects an illustrative default. Deployment-specific priorities may yield different model rankings, as shown in the weight sensitivity analysis (Table A3). Future work should validate these metrics on larger, domain-specific benchmarks with step-level annotated reasoning traces.